

Bimodal Spread Majorization: Bridging Inequality, Bipolarization, and Polarization in Income Tax Policy*

Oriol Carbonell-Nicolau[†] Humberto Llavador[‡]

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Abstract

This paper explores the relationship between income inequality, bipolarization, and polarization using the Bimodal Spread Majorization (BSM) axiom, a novel criterion, weaker than the Lorenz criterion, offering a unified framework for evaluating distributional income divergence. Embedding BSM within the Mirrlees model, we find that inequality-reducing taxation is equivalent to bipolarization-reducing taxation, whereas polarization represents a stricter criterion due to its sensitivity to trade-offs between spread reductions and group identification. Furthermore, marginal-rate progressivity emerges as a universal feature of spread-reducing tax schedules. Finally, we provide characterizations of tax policies that reduce inequality, bipolarization, and polarization.

Keywords: Bimodal Spread Majorization, Income Inequality, Income Polarization, Progressive Taxation, Social Pre-order, Lorenz Criterion, Endogenous Income.

JEL classifications: D31, D63, E62, H23, H24, D71.

1 Introduction

The study of income inequality and polarization has garnered significant attention from economists, policymakers, and social scientists alike. While both phenomena relate to the distribution of income and are perceived as being normatively regrettable, there are conceptual differences between them.

Inequality refers to the overall uneven distribution of resources among individuals or groups and is typically measured by Lorenz-consistent metrics (Lambert, 2002; Chakravarty, 2009, 2015; Cowell, 2011; Myles, 2012).

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[†]Department of Economics, Rutgers University, 75 Hamilton Street, New Brunswick, NJ 08901, USA. E-mail: carbonell-nicolau@rutgers.edu.

[‡]Universitat Pompeu Fabra (UPF) and Barcelona School of Economics (BSE), R. Trias Fargas 25–27, 08005 Barcelona, Spain. E-mail: humberto.llavador@upf.edu. Humberto Llavador acknowledges financial support from the Spanish Research Agency (AEI) through the Severo Ochoa Programme for Centres of Excellence in R&D (Barcelona School of Economics CEX2019-000915-S) funded by MCIN/AEI/10.13039/501100011033 and the grants PID2022-138443NB-I00 and PID2023-153318NB-I00.

Bipolarization, on the other hand, is concerned with the emergence of two distinct groups with a diminishing middle class, reflecting a growing divide between the rich and poor (Wolfson, 1994; Deutsch et al., 2013; Chakravarty, 2015).¹ According to Wolfson (1994), a more bipolarized income distribution is one that is more spread out from the middle so that there are fewer individuals with middle-level incomes. Usually, this spreading out goes hand in hand with a tendency towards bimodality. This is because a smaller middle class is associated with greater separateness of the bottom and top halves of the income distribution and with greater distances between groups.

Finally, polarization encompasses a broader phenomenon where society fragments into multiple distinct clusters or extremes, not limited to just two groups, leading to multipolar social and political division (Esteban and Ray, 1994; Duclos et al., 2004). Hence, conceptually, polarization does not necessarily capture the size of the middle class and is distinct from the notion of bipolarization.²

The literature on polarization is closely linked to the literature on conflict (Reynal-Querol, 2002; Montalvo and Reynal-Querol, 2005; Esteban et al., 2012), and the measures of polarization set forth in the literature (such as those in Esteban and Ray (1994) and Duclos, Esteban and Ray (2004)) rely on the concepts of identification and alienation. Members of each cluster feel a stronger sense of identification with their group the larger the share of the population it represents, while the perceived alienation from other groups increases with the distance between groups. Polarization increases with both identification and alienation. Thus, while inequality focuses on general disparities, bipolarization points to the erosion of the middle class, and polarization measures the clustering around various extremes. Each phenomenon poses unique challenges to societal cohesion and stability.

While inequality, bipolarization, and polarization have been extensively studied as separate phenomena, less attention has been devoted to frameworks that analyze these measures of income spread in an integrated manner. This paper bridges this gap by introducing the Bimodal Spread Majorization (BSM) axiom.

The BSM axiom offers a simple criterion for evaluating income divergence. It only compares perfectly bimodal income distributions—where incomes cluster around two distinct points—with the same total income, and merely requires ranking them according to the degree of divergence between their two income poles. In this context, a greater distance between the poles indicates a more spread income distribution.

The BSM axiom is weaker than the Lorenz criterion, as it only applies to two-group income distributions with the same total income; furthermore, it is satisfied by social pre-orders based on general measures of inequality, bipolarization, and polarization, whether in their relative or absolute forms.

Importantly, this axiom has substantive practical implications beyond its wide-ranging scope. By embedding the BSM axiom within a broad social pre-order framework, we provide a normative foundation for progressivity grounded in principles of equality and depolarization. Analyzing inequality and polarization measures through the lens of the BSM axiom yields valuable insights into “spread-reducing” nonlinear income taxation. This perspective reveals how nonlinear taxation can enhance “social welfare,” as defined by a social pre-order satisfying the BSM axiom, regardless of the initial wage distribution. The analysis is framed within the classical Mirrlees (1971) model, which provides a robust foundation for the study of nonlinear income taxation with endogenous labor supply.

¹In fact, the literature on bipolarization developed hand in hand with the continuous decline in the size of the middle class in the US and the UK since the 1980s (Wolfson, 1994; Jenkins, 1995).

²For a detailed comparison of polarization measures, see Esteban and Ray (2012), Duclos and Taptué (2015) and Chakravarty (2009, 2015).

The relationship between income inequality and taxation has been extensively examined in the economic literature. The seminal contributions of [Jakobsson \(1976\)](#), [Fellman \(1976\)](#), and [Kakwani \(1977\)](#) established the foundational principle that average-rate progressive income taxes are necessary and sufficient for a post-tax income distribution to Lorenz dominate, in the relative sense, any pre-tax income distribution.³ In recent years, this result has been extended to accommodate the endogenous nature of income. [Carbonell-Nicolau and Llavador \(2018, 2021a\)](#) demonstrated that marginal-rate progressivity—i.e., increasing marginal tax rates on income—fully characterizes, under general conditions on preferences, the inequality-reducing principle based on the relative Lorenz dominance criterion. That is, there exists a general class of utility functions for which income taxes are inequality-reducing if and only if they are marginal-rate progressive.

In this paper, we broaden the analysis of spread-reducing income taxation through the lens of the BSM axiom, enabling us to identify common properties of tax schedules that effectively mitigate both inequality and polarization, regardless of the specific metrics used to quantify these phenomena.

Our analysis emphasizes three main findings. First, inequality-reducing taxation is equivalent to bipolarization-reducing taxation, whereas polarization represents a stricter criterion. Specifically, tax-induced polarization mitigation necessarily leads to inequality and bipolarization alleviation, but the converse assertion is generally false. This distinction arises from the inherent ambiguity of polarization measures in resolving trade-offs between spread reductions and increases in identification. In an endogenous income framework, progressive taxation compresses income gaps, thereby reducing inequality while simultaneously causing pre-tax incomes to “bunch” around the thresholds between tax brackets (where marginal tax rates change), increasing identification. Consequently, only proportional tax schedules, which entirely avoid income clustering, can simultaneously reduce both inequality and polarization.

Second, despite these differences, our analysis establishes a general result: spread-reducing tax schedules must exhibit marginal-rate progressivity, characterized by non-decreasing marginal tax rates on income. This result holds consistently across measures of inequality, bipolarization, and polarization.

Finally, building on the equivalences between the various metrics, we fully characterize spread-reducing tax policies for each normative criterion: inequality, bipolarization, and polarization. While instances of these characterizations were obtained in [Carbonell-Nicolau and Llavador \(2021b\)](#) for the specific relative Lorenz and Foster-Wolfson pre-orders, we offer generalizations that encompass any extensions of these pre-orders. Additionally, we broaden the analysis by incorporating novel characterizations for all absolute measures of inequality and bipolarization, as well as both relative and absolute polarization measures.

In conclusion, our findings highlight the critical role of tax progressivity in the design of tax policies that effectively and simultaneously increase equality and depolarization, contributing to a broader discourse on equitable economic policy.

The paper is structured as follows. [Section 2](#) introduces the Bimodal Spread Majorization axiom (BSM) and the concept of a social pre-order as a tool for comparing income distributions. It shows that a wide range of established inequality and polarization measures can be viewed as specific instances of social pre-orders satisfying the BSM axiom. [Section 3](#)

³Most of the extensions of the seminal Jakobsson-Fellman-Kakwani result are framed in terms of exogenous income (see, e.g., [Hemming and Keen \(1983\)](#); [Eichhorn et al. \(1984\)](#); [Liu \(1985\)](#); [Formby et al. \(1986\)](#); [Thon \(1987\)](#); [Latham \(1988\)](#); [Thistle \(1988\)](#); [Moyes \(1988, 1994\)](#); [Le Breton et al. \(1996\)](#); [Ebert and Moyes \(2000\)](#); [Ju and Moreno-Ternero \(2008\)](#)). The first papers introducing the disincentive effects of taxation tended to emphasize negative results ([Allingham, 1979](#); [Onrubia et al., 2005](#); [Ebert and Moyes, 2003, 2007](#)).

examines these measures through the framework of social pre-orders satisfying the BSM axiom, deriving widely applicable insights into “spread-reducing” nonlinear income taxation in the context of endogenous income. Finally, [Section 4](#) concisely summarizes the main results and outlines potential avenues for future research.

2 Social Pre-orders and Bimodal Spread Majorization

This section introduces the Bimodal Spread Majorization (BSM) axiom as a fundamental tool for assessing income dispersion. This axiom provides a natural criterion for ranking perfectly bimodal distributions—those in which two distinct income levels each account for 50% of the population—according to the divergence between these income groups. By identifying social pre-orders (incomplete binary relations over the set of all income distributions) that adhere to the BSM axiom, we are capturing the minimal essence of income spread. We show that all established inequality and polarization measures, both in their relative and their absolute versions, can be interpreted as specific instances of social pre-orders satisfying the BSM axiom.

We begin by defining social pre-orders as incomplete rankings of income distributions.

An **income distribution** is a vector $\mathbf{z} = (z_1, \dots, z_n)$ in $\mathbb{R}_{++}^n$ with its coordinates arranged in non-decreasing order, i.e., $z_1 \leq \dots \leq z_n$; hence, for each i , z_i represents the income of the i th individual, and n is a fixed but otherwise arbitrary even natural number representing the size of the population.⁴ The set of all income distributions is denoted by $\mathcal{Z}_n$. Let $\succsim$ on $\mathcal{Z}_n \times \mathcal{Z}_n$ represent a binary relation on $\mathcal{Z}_n$.

Definition 1. A **social pre-order** is a reflexive and transitive binary relation on $\mathcal{Z}_n$.

We are interested in social pre-orders that rank income distributions based on their spread. We denote this relationship as “ $\mathbf{z} \succsim \mathbf{z}'$ ”, meaning that the income distribution $\mathbf{z}$ exhibits no less dispersion than $\mathbf{z}'$. The asymmetric and symmetric parts of $\succsim$ are denoted by $>$ and $\sim$, respectively.

We take a minimalist approach to income dispersion and propose a novel axiom grounded in the comparative analysis of idealized bimodal income distributions.

An income distribution $\mathbf{z} = (z_1, \dots, z_n)$ in $\mathcal{Z}_n$ is said to be **perfectly bimodal** if

$$\underline{z} := z_1 = \dots = z_{\frac{n}{2}} < z_{\frac{n}{2}+1} = \dots = z_n =: \bar{z}.$$

We sometimes denote $\mathbf{z}$ by $(\underline{z}, \bar{z})$, which is a slight abuse of notation

We now introduce the fundamental BSM axiom.

Bimodal Spread Majorization (BSM). Let $\succsim$ be a binary relation on $\mathcal{Z}_n$. Suppose that $(\underline{x}, \bar{x})$ and $(\underline{y}, \bar{y})$ are perfectly bimodal income distributions with the same total income. Then

$$(\underline{y}, \bar{y}) \succsim (\underline{x}, \bar{x}) \Rightarrow \bar{y} - \underline{y} \geq \bar{x} - \underline{x}.$$

The **BSM** axiom embodies a fundamental and intuitively compelling concept that underpins a wide spectrum of normative criteria for evaluating income distributions. In what follows, we demonstrate that the axiom encompasses all existing measures of inequality and polarization. Moreover, due to its abstract formulation and inherent simplicity, the axiom’s reach extends beyond these conventional frameworks.

Our analysis covers both relative and absolute measures of inequality and polarization, characterized by the Scale Invariance and Translation Invariance axioms, respectively. These fundamental invariance properties, well-established in the literature, can be formally stated as follows:

⁴We omit the analysis for the case when n is odd, which is similar but requires its own separate terminology.

Scale Invariance (SI). Suppose that $\succsim$ is a social pre-order. If $\mathbf{z}'$ is obtained from $\mathbf{z} \in \mathcal{Z}_n$ by a proportional change in incomes, then $\mathbf{z}' \sim \mathbf{z}$.

Translation Invariance (TI). Suppose that $\succsim$ is a social pre-order. If $\mathbf{z}' \in \mathcal{Z}_n$ is obtained from $\mathbf{z} \in \mathcal{Z}_n$ by a translation in incomes, then $\mathbf{z}' \sim \mathbf{z}$.

Accordingly, we classify social pre-orders based on their invariance properties: *relative pre-orders*, which satisfy scale invariance (SI), and *absolute pre-orders*, which satisfy translation invariance (TI).

2.1 Inequality

A fundamental tenet in the literature on inequality is that its measurement should adhere to the “Transfer Principle.”

A distribution $\mathbf{z}'$ is obtained from $\mathbf{z} \in \mathcal{Z}_n$ by a **progressive transfer** if $\mathbf{z}'$ results from a transfer of income from one individual to anyone poorer up to the point where both individuals have the same income.

In this section, we apply the notion of social pre-order to a general class of inequality measures based solely on this principle. Formally:

Transfer Principle (TP) (Pigou, 1912; Dalton, 1920). Suppose that $\succsim$ is a reflexive and transitive binary relation on $\mathcal{Z}_n$. If $\mathbf{z}'$ is obtained from $\mathbf{z} \in \mathcal{Z}_n$ by a progressive transfer, then $\mathbf{z} \succ \mathbf{z}'$.

We define an inequality pre-order as a binary relation satisfying the Transfer Principle.

Definition 2. An **inequality pre-order** $\succsim_I$ is a social pre-order satisfying TP, with “ $\mathbf{z} \succsim_I \mathbf{z}'$ ” understood as “ $\mathbf{z}$ is (weakly) more unequal than $\mathbf{z}'$.”

Because TP is stronger than BSM, an inequality pre-order is a particular instance of a social pre-order satisfying BSM.

Proposition 1. Any inequality pre-order satisfies BSM.

Proof. Suppose that $(\underline{x}, \bar{x})$ and $(\underline{y}, \bar{y})$ are perfectly bimodal income distributions with the same total income. Suppose further that $(\underline{y}, \bar{y}) \succsim_I (\underline{x}, \bar{x})$. We must show that

$$\bar{y} - \underline{y} \geq \bar{x} - \underline{x}. \quad (1)$$

This inequality is clearly true if $(\underline{x}, \bar{x}) = (\underline{y}, \bar{y})$, so suppose that $(\underline{x}, \bar{x}) \neq (\underline{y}, \bar{y})$. Then, since $(\underline{x}, \bar{x})$ and $(\underline{y}, \bar{y})$ have the same total income, it follows that

$$\text{either } [\underline{x} < \underline{y} \ \& \ \bar{x} > \bar{y}] \text{ or } [\underline{x} > \underline{y} \ \& \ \bar{x} < \bar{y}].$$

In the first case, $(\underline{y}, \bar{y})$ can be obtained from $(\underline{x}, \bar{x})$ by $n/2$ progressive transfers, implying (by TP) that $(\underline{x}, \bar{x}) \succ_I (\underline{y}, \bar{y})$, a contradiction.⁵ Hence,

$$\underline{x} > \underline{y} \quad \text{and} \quad \bar{x} < \bar{y},$$

which yields (1). ■

⁵Observe that $(\underline{x}, \bar{x}) \succ_I (\underline{y}, \bar{y})$ implies that $(\underline{x}, \bar{x}) \succsim_I (\underline{y}, \bar{y})$, which, combined with the relation $(\underline{y}, \bar{y}) \succsim_I (\underline{x}, \bar{x})$, gives $(\underline{x}, \bar{x}) \sim_I (\underline{y}, \bar{y})$. This contradicts the fact that $(\underline{x}, \bar{x}) \succ_I (\underline{y}, \bar{y})$.

2.2 Bipolarization

Bipolarization serves as a formal metric for measuring the decline of the middle class. This concept rests on two fundamental axioms in the literature: the “Increased Spread” and “Increased Bipolarity” axioms (Foster and Wolfson, 2010; Chakravarty, 2015). These axioms provide a foundation for understanding how income distributions become more bipolarized, either through widening gaps between two income groups or through clustering of incomes around two distinct poles.

Prior to defining a bipolarization pre-order, we need some preliminary notation.

For any $\mathbf{z} \in \mathcal{Z}_n$, let

$$z_m := \frac{z_{\frac{n}{2}} + z_{\frac{n}{2}+1}}{2}, \quad \mathbf{z}_- = (z_1, \dots, z_{\frac{n}{2}}), \quad \text{and} \quad \mathbf{z}_+ = (z_{\frac{n}{2}+1}, \dots, z_n).$$

Let “ $\mathbf{z} < \mathbf{z}'$ ” mean that “ $z_i \leq z'_i$ for all i with at least one strict inequality,”⁶ and let “ $\mathbf{z}' T \mathbf{z}$ ” represent the condition that $\mathbf{z}'$ is obtained from $\mathbf{z}$ by a progressive transfer.

We now define the two standard axioms for bipolarization measurement.

Increased Spread (IS) (Foster and Wolfson (2010); Chakravarty (2015)). Let $\succsim$ be a binary relation on $\mathcal{Z}_n$. If $\mathbf{x}$ and $\mathbf{y}$ have the same median ($x_m = y_m$) and

$$\begin{aligned} &[\mathbf{x}_- = \mathbf{y}_- \text{ and } \mathbf{y}_+ < \mathbf{x}_+] \text{ or} \\ &[\mathbf{x}_- < \mathbf{y}_- \text{ and } \mathbf{y}_+ = \mathbf{x}_+] \text{ or} \\ &[\mathbf{x}_- < \mathbf{y}_- \text{ and } \mathbf{y}_+ < \mathbf{x}_+], \text{ then } \mathbf{x} > \mathbf{y}. \end{aligned}$$

Increased Bipolarity (IB) (Foster and Wolfson (2010); Chakravarty (2015)). Let $\succsim$ be a binary relation on $\mathcal{Z}_n$. If $\mathbf{x}$ and $\mathbf{y}$ have the same median ($x_m = y_m$) and

$$\begin{aligned} &[\mathbf{x}_- = \mathbf{y}_- \text{ and } \mathbf{x}_+ T \mathbf{y}_+] \text{ or} \\ &[\mathbf{x}_- T \mathbf{y}_- \text{ and } \mathbf{x}_+ = \mathbf{y}_+] \text{ or} \\ &[\mathbf{x}_- T \mathbf{y}_- \text{ and } \mathbf{x}_+ T \mathbf{y}_+], \text{ then } \mathbf{x} > \mathbf{y}. \end{aligned}$$

Definition 3. A **bipolarization pre-order** $\succsim_B$ is a social pre-order satisfying **IS** and **IB**, with “ $\mathbf{z}' \succsim_B \mathbf{z}$ ” understood as “ $\mathbf{z}'$ is (weakly) more bipolarized than $\mathbf{z}$.”

Bipolarization pre-orders are also special cases of social pre-orders satisfying **BSM**. In fact, the **IS** axiom is in itself already stronger than **BSM**.⁷ Therefore, any binary relation satisfying **IS** also satisfies **BSM**.

Proposition 2. Let $\succsim$ be a reflexive and transitive binary relation on $\mathcal{Z}_n$ satisfying **IS**. Then $\succsim$ satisfies **BSM**.

Proof. Suppose that $(\underline{x}, \bar{x})$ and $(\underline{y}, \bar{y})$ are perfectly bimodal income distributions with the same total income. Suppose further that $(\underline{y}, \bar{y}) \succsim (\underline{x}, \bar{x})$. We must show that

$$\bar{y} - \underline{y} \geq \bar{x} - \underline{x}. \quad (2)$$

This inequality is clearly true if $(\underline{x}, \bar{x}) = (\underline{y}, \bar{y})$, so suppose that $(\underline{x}, \bar{x}) \neq (\underline{y}, \bar{y})$. Then, since $(\underline{x}, \bar{x})$ and $(\underline{y}, \bar{y})$ have the same total income, it follows that

$$\text{either } [\underline{x} < \underline{y} \text{ \& } \bar{x} > \bar{y}] \text{ or } [\underline{x} > \underline{y} \text{ \& } \bar{x} < \bar{y}].$$

⁶We write “ $\mathbf{x} \gg \mathbf{y}$ ” to represent the case when $x_i > y_i$ for all i ; “ $\mathbf{x} > \mathbf{y}$ ” if $x_i \geq y_i$ for all i and $x_j > y_j$ for at least one j ; and “ $\mathbf{x} \geq \mathbf{y}$ ” if $x_i \geq y_i$ for all i , i.e., either $\mathbf{x} > \mathbf{y}$ or $\mathbf{x} = \mathbf{y}$.

⁷The Between-Group Spread axiom in Bossert and Schworm (2008) is simply **IS** without the restriction to equal medians, and hence also stronger than **BSM**.

In the first case, **IS** gives $(\underline{x}, \bar{x}) \succ (\underline{y}, \bar{y})$, a contradiction.⁸ Hence,

$$\underline{x} > \underline{y} \quad \text{and} \quad \bar{x} < \bar{y},$$

which yields (2). ■

2.3 Polarization

The literature on polarization has largely focused on the polarization index introduced by **Esteban and Ray (1994)**, which offers a distinct approach to measuring economic polarization. Unlike bipolarization metrics that emphasize the decline of the middle class, the Esteban–Ray measure captures the concept of multipolarity in income distributions, quantifying how populations cluster around multiple income levels. Despite this fundamental difference in focus, the Esteban–Ray measure also satisfies the **BSM** axiom—the same principle underlying inequality and bipolarization measures.

The *relative* version of the index takes the following form:

$$P_R(\mathbf{z}) = K \sum_{i=1}^n \sum_{j=1}^n \pi_{\mathbf{z}}(z_i)^\alpha |\ln z_i - \ln z_j|, \quad (3)$$

where $\pi_{\mathbf{z}}(z)$ denotes the number of individuals whose income is z , and $K > 0$ and $\alpha \in (0, 1.6]$ are parameters.⁹

The *absolute* counterpart of P_R , P_A , is given by

$$P_A(\mathbf{z}) = K \sum_{i=1}^n \sum_{j=1}^n \pi_{\mathbf{z}}(z_i)^\alpha |z_i - z_j|.$$

The associated relative and absolute polarization orders, respectively denoted by $\succ_{RP}$ and $\succ_{AP}$, are defined on $\mathcal{X}_n$ by

$$\mathbf{x} \succ_{RP} \mathbf{y} \Leftrightarrow P_R(\mathbf{x}) \geq P_R(\mathbf{y}) \quad \text{and} \quad \mathbf{x} \succ_{AP} \mathbf{y} \Leftrightarrow P_A(\mathbf{x}) \geq P_A(\mathbf{y}).$$

Consequently, unlike the inequality and bipolarization pre-orders, the polarization order is complete.

We establish the alignment of the polarization orders with the **BSM** through the following two results.

Proposition 3. *The relative polarization order $\succ_{RP}$ is scale-invariant and satisfies **BSM**.*

Proof. First, we show that $\succ_{RP}$ is scale-invariant. Let $\mathbf{z}, \mathbf{z}' \in \mathcal{X}_n$ such that $\mathbf{z}' = \beta \mathbf{z}$ for some $\beta > 0$. Then,

$$P_R(\mathbf{z}') = K \sum_{i=1}^n \sum_{j=1}^n \pi_{\mathbf{z}'}(z'_i)^\alpha |\ln z'_i - \ln z'_j| = K \sum_{i=1}^n \sum_{j=1}^n \pi_{\mathbf{z}}(z_i)^\alpha |\ln z_i - \ln z_j| = P_R(\mathbf{z}).$$

To see that **BSM** is satisfied, suppose that $(\underline{x}, \bar{x})$ and $(\underline{y}, \bar{y})$ are perfectly bimodal income distributions with the same total income. We have

$$\begin{aligned} (\underline{y}, \bar{y}) \succ_{RP} (\underline{x}, \bar{x}) &\Leftrightarrow P_R(\underline{y}, \bar{y}) \geq P_R(\underline{x}, \bar{x}) \Leftrightarrow K \left(\frac{n}{2}\right)^{2+\alpha} 2(\ln \bar{y} - \ln \underline{y}) \geq K \left(\frac{n}{2}\right)^{2+\alpha} 2(\ln \bar{x} - \ln \underline{x}) \\ &\Leftrightarrow \ln \bar{y} - \ln \underline{y} \geq \ln \bar{x} - \ln \underline{x} \Leftrightarrow \frac{\bar{y}}{\underline{y}} \geq \frac{\bar{x}}{\underline{x}}. \end{aligned}$$

⁸ $(\underline{x}, \bar{x}) \succ (\underline{y}, \bar{y})$ implies that $(\underline{x}, \bar{x}) \succ (\underline{y}, \bar{y})$, which, combined with the relation $(\underline{y}, \bar{y}) \succ (\underline{x}, \bar{x})$, gives $(\underline{x}, \bar{x}) \sim (\underline{y}, \bar{y})$. This contradicts the fact that $(\underline{x}, \bar{x}) \succ (\underline{y}, \bar{y})$.

⁹We show in Appendix A that our formulation is equivalent to the expression in **Esteban and Ray (1994)**.

Because both distributions have the same total income (i.e., $\bar{y} + \underline{y} = \bar{x} + \underline{x}$), we have

$$\bar{y} \geq \bar{x} \geq \underline{x} \geq \underline{y},$$

whence $\bar{y} - \underline{y} \geq \bar{x} - \underline{x}$. Hence, **BSM** holds. $\blacksquare$

Proposition 4. *The absolute polarization order $\succ_{AP}$ is translation-invariant and satisfies **BSM**.*

Proof. First, we show that $\succ_{AP}$ is translation-invariant. Let $\mathbf{z}, \mathbf{z}' \in \mathcal{Z}_n$ such that $\mathbf{z}' = \mathbf{z} + \beta$. Then,

$$P_A(\mathbf{z}') = K \sum_{i=1}^n \sum_{j=1}^n \pi_{\mathbf{z}'}(z'_i)^\alpha |z'_i - z'_j| = K \sum_{i=1}^n \sum_{j=1}^n \pi_{\mathbf{z}}(z_i)^\alpha |z_i - z_j| = P_A(\mathbf{z}).$$

To see that **BSM** is satisfied, suppose that $(\underline{x}, \bar{x})$ and $(\underline{y}, \bar{y})$ are perfectly bimodal income distributions with the same total income. We have

$$(\underline{y}, \bar{y}) \succ_{AP} (\underline{x}, \bar{x}) \Leftrightarrow P_A(\underline{y}, \bar{y}) \geq P_A(\underline{x}, \bar{x}) \Leftrightarrow K \left(\frac{n}{2}\right)^{2+\alpha} 2(\bar{y} - \underline{y}) \geq K \left(\frac{n}{2}\right)^{2+\alpha} 2(\bar{x} - \underline{x}) \Leftrightarrow \bar{y} - \underline{y} \geq \bar{x} - \underline{x}.$$

Hence, **BSM** holds. $\blacksquare$

3 Spread-Reducing Income Taxation

Having established the broad scope of the **BSM** axiom, we now turn our attention to the analysis of spread-reducing income taxation. This concept refers to tax systems that reduce income dispersion in a general sense, which can be interpreted as reducing both inequality and polarization through the tax code, as demonstrated in the previous section.

Our analysis adopts a Mirrleesian framework with endogenous income, focusing on universal spread-reducing conditions that are independent of specific initial wage distributions.

The following subsections introduce the concept of a tax schedule and outline our analytical framework. Subsequently, we present a comprehensive characterization of spread-reducing income tax schedules.

3.1 Tax schedules

A tax schedule is a function that assigns a tax liability to each gross income level. Our analysis focuses on the class of tax schedules that are continuous, non-decreasing, order-preserving, and piecewise linear. Formally,

Definition 4. Define a $(K+1)$ -**bracket piecewise linear tax schedule** as a continuous and nondecreasing map $T: \mathbb{R}_+ \rightarrow \mathbb{R}$ that assigns to each income level $y \in \mathbb{R}_+$ a tax liability

$$T(y) := \begin{cases} -\alpha_0 + t_0 y & \text{if } 0 = \bar{y}_0 \leq y \leq \bar{y}_1, \\ -\alpha_0 + t_0 \bar{y}_1 + t_1(y - \bar{y}_1) & \text{if } \bar{y}_1 < y \leq \bar{y}_2, \\ \vdots & \vdots \\ -\alpha_0 + t_0 \bar{y}_1 + t_1(\bar{y}_2 - \bar{y}_1) + \cdots + t_{K-1}(\bar{y}_K - \bar{y}_{K-1}) + t_K(y - \bar{y}_K) & \text{if } \bar{y}_K < y, \end{cases}$$

where $\alpha_0 \geq 0$, $K \in \mathbb{Z}_+$, $t_k \in [0, 1)$ for each $k \in \{0, \dots, K\}$, $t_k \neq t_{k+1}$ whenever $k \in \{0, \dots, K-1\}$ and $K \geq 1$, and $0 = \bar{y}_0 < \cdots < \bar{y}_K$.

For $K \in \mathbb{Z}_+$, the set of $(K+1)$ -bracket piecewise linear tax schedules is denoted by $\mathcal{T}_K$, and the set of all piecewise linear tax schedules is defined as

$$\mathcal{T} := \bigcup_{K=1}^{\infty} \mathcal{T}_K.$$

The following subclasses of $\mathcal{T}$ will play an important role in our analysis.

- Let $\mathcal{T}_{lin}$ denote the set of all **linear tax schedules** in $\mathcal{T}$, where each linear tax schedule is defined by $T(y) = -\alpha + ty$ for all $y \in \mathbb{R}_+$, with parameters $\alpha \geq 0$ and $t \in [0, 1)$.
- Let $\mathcal{T}_{prop}$ denote the set of all **proportional tax schedules** in $\mathcal{T}$, which take the form $T(y) = ty$ for some $t \in [0, 1)$.
- Let $\mathcal{T}_{m-prog}$ denote the set of all **marginal-rate progressive tax schedules** in $\mathcal{T}$, where a tax schedule $T \in \mathcal{T}$ is marginal-rate progressive if it is a convex function. In words, a tax schedule is marginal-rate progressive if it exhibits non-decreasing marginal tax rates on income.
- Given $b \geq 0$ and $R \subseteq [0, 1)$, let

$$\mathcal{T}(b, R) := \{(\alpha_0, \mathbf{t}, \bar{\mathbf{y}}) \in \mathcal{T} : \alpha_0 \geq b \text{ and } t_k \in R \text{ for all } k\}.$$

In words, $\mathcal{T}(b, R)$ is the set of all piecewise linear tax schedules that endow all agents with a subsidy of at least b and whose marginal tax rates t_k ($k = 0, 1, \dots$) are contained in the subset R . Note that for $b = 0$ and $R = [0, 1)$, $\mathcal{T}(b, R)$ is simply the set of all piecewise linear tax schedules, $\mathcal{T}$.

- Finally, denote the intersection $\mathcal{T}(b, R) \cap \mathcal{T}_{m-prog}$ by $\mathcal{T}_{m-prog}(b, R)$ and the intersection $\mathcal{T}(b, R) \cap \mathcal{T}_{lin}$ by $\mathcal{T}_{lin}(b, R)$.

3.2 Endogenous Income Framework

To allow for potential disincentive effects of taxation on work effort, we adopt the standard Mirrlees model (Mirrlees, 1971). The description of the setup follows Carbonell-Nicolau and Llavador (2021a) closely.

The utility function $u : \mathbb{R}_+ \times [0, 1] \rightarrow \mathbb{R}$, defined over consumption-labor pairs $(c, l) \in \mathbb{R}_+ \times [0, 1]$, is assumed to be continuous, with $u(\cdot, l)$ strictly increasing in c for each $l \in [0, 1)$ and $u(c, \cdot)$ strictly decreasing in l for each $c > 0$. Additionally, u is assumed to be strictly quasiconcave on $\mathbb{R}_{++} \times [0, 1)$ and twice continuously differentiable on $\mathbb{R}_{++} \times (0, 1)$. Finally, we assume that for each $a > 0$, there exists some l such that $u(al, l) > u(\mathbf{0})$.¹⁰

For $(c, l) \in \mathbb{R}_{++} \times (0, 1)$, let

$$MRS(c, l) := -\frac{u_l(c, l)}{u_c(c, l)}$$

denote the marginal rate of substitution of labor for consumption, where

$$u_c(c, l) := \frac{\partial u(c, l)}{\partial c} \quad \text{and} \quad u_l(c, l) := \frac{\partial u(c, l)}{\partial l}.$$

The following standard assumptions will be maintained for each $c > 0$:

$$\lim_{l \rightarrow 1^-} MRS(c, l) = +\infty \quad \text{and} \quad \lim_{l \rightarrow 0^+} MRS(c, l) < +\infty. \quad (4)$$

The first condition states that the compensation required by an individual for an extra unit of working time tends to infinity as the agent's leisure time approaches zero. The second condition is a mild finiteness condition on the marginal rate of substitution of labor for consumption.

An **ability distribution** is a vector $\mathbf{a} = (a_1, \dots, a_n)$ in $\mathbb{R}_{++}^n$ such that the coordinates in $\mathbf{a}$ are arranged in increasing order, i.e., $a_1 \leq \dots \leq a_n$; hence, for each i , a_i represents the ability

¹⁰This last assumption ensures that individuals consume a positive amount in the absence of subsidies.

level of the i th agent. Let $\mathcal{A}_n$ represent the set of all ability distributions with population size n .

An agent of ability $a > 0$ who chooses $l \in [0, 1]$ units of labor and faces a tax schedule $T \in \mathcal{T}$ consumes $c = al - T(al)$ units of the good and derives a utility of $u(c, l)$. Thus, the agent's problem is

$$\max_{l \in [0, 1]} u(al - T(al), l). \quad (5)$$

A **solution function** is a map $l^u : \mathbb{R}_{++} \times \mathcal{T} \rightarrow [0, 1]$ such that $l^u(a, T)$ is a solution to (5) for each $(a, T) \in \mathbb{R}_{++} \times \mathcal{T}$. A solution function l^u induces **pre-tax** and **post-tax income functions** $y^u : \mathbb{R}_{++} \times \mathcal{T} \rightarrow \mathbb{R}_+$ and $x^u : \mathbb{R}_{++} \times \mathcal{T} \rightarrow \mathbb{R}_+$, respectively, defined by

$$y^u(a, T) := al^u(a, T) \quad \text{and} \quad x^u(a, T) := al^u(a, T) - T(al^u(a, T)).$$

We write $y^u(a, 0)$ and $x^u(a, 0)$ to denote, respectively, the pre-tax and post-tax incomes of an a -type in the absence of taxation, i.e., when the tax schedule is identically zero ($T \equiv 0$).

Given $a > 0$, let $U^a : \mathbb{R}_+ \times [0, a] \rightarrow \mathbb{R}$ be defined by $U^a(c, y) := u(c, y/a)$. For $(c, y, a) \in \mathbb{R}_{++}^3$ with $y < a$, define

$$\eta^a(c, y) := -\frac{\partial U^a(c, y)}{\partial y} \bigg/ \frac{\partial U^a(c, y)}{\partial c}.$$

The following is the standard *agent monotonicity* condition introduced by [Mirrlees \(1971\)](#) (see also [Seade \(1982\)](#) and [Myles \(1995, p. 136\)](#)).

Definition 5. A utility function satisfies **agent monotonicity** if $\eta^a(c, y) > \eta^{a'}(c, y)$ for each $(c, y) \in \mathbb{R}_+^2$ and $0 < a < a'$ with $y < a$.¹¹

The set of utility functions satisfying the conditions in (4) and agent monotonicity is denoted by $\mathcal{U}$. This domain contains the standard utility functions used in the literature (see [Carbonell-Nicolau and Llavador, 2018](#)).

For each $(a, b) \in \mathbb{R}_{++} \times \mathbb{R}_+$, consider the problem

$$\max_{l \in [0, 1]} u(al + b, l). \quad (6)$$

This is the problem faced by an a -agent who receives a subsidy b . Since u is strictly quasiconcave on $\mathbb{R}_{++} \times [0, 1]$, for each $(a, b) \in \mathbb{R}_{++} \times \mathbb{R}_+$, there is a unique solution $l^u(a, b)$ to (6). Moreover, for given $b \geq 0$, the derivative of the map $a \mapsto l^u(a, b)$ exists for all but at most one $a > 0$ (see [Carbonell-Nicolau and Llavador, 2018](#)).

3.3 Characterizing Spread-Reducing Tax Schedules

Analyzing spread-reducing income taxation through the lens of the **BSM** axiom enables us to identify common properties of tax schedules that effectively mitigate both inequality and polarization, regardless of the specific metrics used to quantify these phenomena.

Our analysis reveals a significant equivalence between inequality-reducing and bipolarization-reducing taxation, which we demonstrate in detail in the following subsection. However, the relationship between these two metrics and polarization is more nuanced. We find that the general equivalence among inequality, bipolarization, and polarization holds only for a specific subclass of tax schedules—namely, proportional tax schedules. We develop the intuition behind this result in the next subsection.

¹¹[Mirrlees \(1971\)](#) uses the weak version of this monotonicity condition (i.e., $\eta^a(c, y) \geq \eta^{a'}(c, y)$ for each $(c, y) \in \mathbb{R}_+^2$ and $0 < a < a'$ with $y < a$). All our results, except for [Theorem 5](#) and [Theorem 9](#), remain intact under the weaker condition.

We then establish a general principle: spread-reducing tax schedules must exhibit marginal-rate progressivity, i.e., non-decreasing marginal tax rates on income.

Building on the necessity of marginal-rate progressivity and on the equivalences between the various metrics discussed in the next subsection, the final subsection presents a comprehensive characterization of spread-reducing taxes, tailored to the specific measures of inequality and polarization under consideration.

We focus on the characterization of tax structures inducing spread-reducing post-tax income distributions, relative to the tax-free distribution, with respect to a social pre-order.

Definition 6. Given $u \in \mathcal{U}$ and a social pre-order $\succsim$, a tax schedule $T \in \mathcal{T}$ is $(\succsim, u)$ -**reducing**, denoted by $(\succsim, u)$ - r , if

$$(y^u(a_1, 0), \dots, y^u(a_n, 0)) \succsim (x^u(a_1, T), \dots, x^u(a_n, T))$$

for every ability distribution $(a_1, \dots, a_n) \in \mathcal{A}_n$ and every pre-tax and post-tax income functions y^u and x^u induced by u .

We can view $(\succsim, u)$ -reducing tax schedules as those leading to a less dispersed income distribution relative to the tax-free distribution according to the specific social pre-order. For $u \in \mathcal{U}$, denote by $\mathcal{T}_{(\succsim, u)\text{-}r}$ the set of spread-reducing tax schedules relative to any social pre-order $(\succsim, u)$.

An example of such a pre-order is the relative Lorenz pre-order.

To maintain notational consistency with our general framework—where “greater” elements in a pre-order indicate greater dispersion—we invert the standard relative Lorenz dominance relation. We denote this inverted pre-order as $\succsim'_{RL}$, which captures relative Lorenz *subordination* (i.e., greater inequality).

Definition 7. Given two income distributions $z = (z_1, \dots, z_n)$ and $z' = (z'_1, \dots, z'_n)$ with $z_n, z'_n > 0$, z is *more unequal than* z' in the relative Lorenz sense, denoted by $z \succsim'_{RL} z'$, if

$$\frac{\sum_{i=1}^k z_i}{\sum_{i=1}^n z_i} \leq \frac{\sum_{i=1}^k z'_i}{\sum_{i=1}^n z'_i}, \quad \text{for all } k \in \{1, \dots, n\}.$$

A first result shows that, for relative pre-orders satisfying **BSM**, $(\succsim, u)$ -reducing tax schedules must also be $(\succsim'_{RL}, u)$ -reducing. This is formalized in the following result.

Theorem 1. For any $u \in \mathcal{U}$ and any social pre-order $\succsim$ satisfying **BSM** and **SI**, $\mathcal{T}_{(\succsim, u)\text{-}r} \subseteq \mathcal{T}_{(\succsim'_{RL}, u)\text{-}r}$.

Proof. Suppose that $T \in \mathcal{T}$ is $(\succsim, u)$ - r . Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathcal{A}_n$. Choose any pair a_i and a_j with $a_i < a_j$. Let $\mathbf{a}' = (a'_1, \dots, a'_n)$ satisfy

$$a'_1 = \dots = a'_{\frac{n}{2}} = a_i < a_j = a'_{\frac{n}{2}+1} = \dots = a'_n.$$

Note that

$$\underline{y} := y^u(a'_1, 0) = \dots = y^u(a'_{\frac{n}{2}}, 0) < y^u(a'_{\frac{n}{2}+1}, 0) = \dots = y^u(a'_n, 0) =: \bar{y}$$

and

$$\underline{x} := x^u(a'_1, T) = \dots = x^u(a'_{\frac{n}{2}}, T) \leq x^u(a'_{\frac{n}{2}+1}, T) = \dots = x^u(a'_n, T) =: \bar{x}.$$

Consequently, the two distributions $(\alpha \underline{y}, \alpha \bar{y})$ and $(\underline{x}, \bar{x})$ satisfy

$$\alpha \underline{y} + \alpha \bar{y} = \underline{x} + \bar{x},$$

where $\alpha := \frac{x+\bar{x}}{y+\bar{y}}$. Since $\succsim$ is scale invariant, we have $(\alpha y, \alpha \bar{y}) \sim (y, \bar{y})$. Because T is $(\succsim, u)$ -r, we have $(y, \bar{y}) \succsim (x, \bar{x})$. Consequently,

$$(\alpha y, \alpha \bar{y}) \sim (y, \bar{y}) \succsim (x, \bar{x}),$$

implying that $(\alpha y, \alpha \bar{y}) \succsim (x, \bar{x})$ (Sen, 2017, Lemma 1*a, p. 56). By BSM, we have

$$\alpha \bar{y} - \alpha y \geq \bar{x} - x.$$

Since $\alpha y + \alpha \bar{y} = x + \bar{x}$, it follows that

$$\alpha y \leq x \quad \text{and} \quad \alpha \bar{y} \geq \bar{x},$$

implying that

$$\frac{x}{\alpha y} \geq 1 \geq \frac{\bar{x}}{\alpha \bar{y}}.$$

Consequently,

$$\frac{x^u(a_i, T)}{y^u(a_i, 0)} = \frac{x}{y} \geq \frac{\bar{x}}{\bar{y}} = \frac{x^u(a_j, T)}{y^u(a_j, 0)}.$$

Since a_i and a_j were arbitrary coordinates in $\mathbf{a}$ with $a_i < a_j$, it follows from Lemma 2 in Carbonell-Nicolau and Llavador (2021b) that T is $(\succsim'_{RL}, u)$ -r.

We have established that $\mathcal{T}_{(\succsim, u)-r} \subseteq \mathcal{T}_{(\succsim'_{RL}, u)-r}$. ■

A similar result can be proven for absolute social pre-orders $\succsim$ satisfying BSM: $(\succsim, u)$ -reducing tax schedules must also be $(\succsim'_{AL}, u)$ -reducing. As with the relative case, $\succsim'_{AL}$ represents the inverted absolute Lorenz pre-order—capturing absolute Lorenz *subordination* (i.e., greater absolute dispersion)—based on the criterion introduced by Moyes (1988).

Definition 8. Given two income distributions $\mathbf{z} = (z_1, \dots, z_n)$ and $\mathbf{z}' = (z'_1, \dots, z'_n)$, $\mathbf{z}$ is more unequal than $\mathbf{z}'$ in the absolute Lorenz sense, denoted by $\mathbf{z} \succsim'_{AL} \mathbf{z}'$, if

$$\sum_{i=1}^k z_i - \frac{k}{n} \sum_{i=1}^n z_i \leq \sum_{i=1}^k z'_i - \frac{k}{n} \sum_{i=1}^n z'_i, \quad \text{for all } k \in \{1, \dots, n\}.$$

Theorem 2. For any $u \in \mathcal{U}$ and a any social pre-order $\succsim$ satisfying BSM and TI, $\mathcal{T}_{(\succsim, u)-r} \subseteq \mathcal{T}_{(\succsim'_{AL}, u)-r}$.

We use the following lemma, whose proof is relegated to Appendix B, to prove Theorem 2. This lemma provides a way to check whether tax schedules are inequality-reducing with respect to the absolute Lorenz criterion.

Lemma 1. For $u \in \mathcal{U}$, a tax schedule $T \in \mathcal{T}$ is $(\succsim'_{AL}, u)$ -r if and only if, for any pre-tax and post-tax income functions y^u and x^u ,

$$y^u(a', 0) - y^u(a, 0) \geq x^u(a', T) - x^u(a, T), \quad \text{whenever } a' > a > 0. \quad (7)$$

Proof of Theorem 2. Suppose that $T \in \mathcal{T}$ is $(\succsim, u)$ -r. Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathcal{A}_n$ (n even). Choose any pair a_i and a_j with $a_i < a_j$. Let $\mathbf{a}' = (a'_1, \dots, a'_n)$ satisfy

$$a'_1 = \dots = a'_{\frac{n}{2}} = a_i < a_j = a'_{\frac{n}{2}+1} = \dots = a'_n.$$

Note that

$$\underline{y} := y^u(a'_1, 0) = \dots = y^u(a'_{\frac{n}{2}}, 0) < y^u(a'_{\frac{n}{2}+1}, 0) = \dots = y^u(a'_n, 0) =: \bar{y}$$

and

$$\underline{x} := x^u(a'_1, T) = \dots = x^u(a'_{\frac{n}{2}}, T) \leq x^u(a'_{\frac{n}{2}+1}, T) = \dots = x^u(a'_n, T) =: \bar{x}.$$

Suppose that $\underline{y} + \bar{y} \geq \underline{x} + \bar{x}$ (the case when $\underline{y} + \bar{y} < \underline{x} + \bar{x}$ can be handled similarly). Define $\varepsilon \geq 0$ by

$$\underline{y} + \bar{y} = \underline{x} + \varepsilon + \bar{x} + \varepsilon. \quad (8)$$

Since $\succsim$ is translation invariant, we have $(\underline{x} + \varepsilon, \bar{x} + \varepsilon) \sim (\underline{x}, \bar{x})$. Because T is $(\succsim, u)$ -r, we have $(\underline{y}, \bar{y}) \succsim (\underline{x}, \bar{x})$. Consequently,

$$(\underline{y}, \bar{y}) \succsim (\underline{x}, \bar{x}) \sim (\underline{x} + \varepsilon, \bar{x} + \varepsilon),$$

implying that $(\underline{y}, \bar{y}) \succsim (\underline{x} + \varepsilon, \bar{x} + \varepsilon)$ (Sen, 2017, Lemma 1*a, p. 56). By (8) and BSM, we have

$$\bar{y} - \underline{y} \geq (\bar{x} + \varepsilon) - (\underline{x} + \varepsilon) = \bar{x} - \underline{x}.$$

Consequently,

$$y^u(a_j, 0) - y^u(a_i, 0) \geq x^u(a_j, T) - x^u(a_i, T).$$

Since a_i and a_j were arbitrary coordinates in $\mathbf{a}$ with $a_i < a_j$, it follows from Lemma 1 that T is $(\succsim'_{AL}, u)$ -r.

We have established the containment $\mathcal{T}_{(\succsim, u)\text{-}r} \subseteq \mathcal{T}_{(\succsim'_{AL}, u)\text{-}r}$. ■

The next section considers inequality pre-orders $\succsim_I$, bipolarization pre-orders $\succsim_B$, and the orders based on the Esteban–Ray polarization index, in their absolute and relative forms, $\succsim_{AP}$ and $\succsim_{RP}$, respectively.

In the context of inequality and bipolarization measures, we will also differentiate between *relative* pre-orders, satisfying scale invariance (SI), and *absolute* pre-orders, satisfying translation invariance (TI). Relative inequality and bipolarization pre-orders are represented as $\succsim_{RI}$ and $\succsim_{RB}$, respectively, while their absolute counterparts are denoted by $\succsim_{AI}$ and $\succsim_{AB}$.

3.3.1 Equivalences

This section explores the interplay between tax-mediated inequality and polarization reduction. Specifically, we wish to compare sets of $(\succsim, u)$ -reducing tax schedules.

First, using Theorem 1 and Theorem 2, we can establish the general equivalence between inequality and bipolarization-reducing tax schedules.

Theorem 3. *For any $u \in \mathcal{U}$ and any two relative inequality and bipolarization pre-orders, $\succsim_{RI}$ and $\succsim_{RB}$,*

$$\mathcal{T}_{(\succsim_{RI}, u)\text{-}r} = \mathcal{T}_{(\succsim_{RB}, u)\text{-}r}. \quad (9)$$

For any $u \in \mathcal{U}$ and any two absolute inequality and bipolarization pre-orders, $\succsim_{AI}$ and $\succsim_{AB}$,

$$\mathcal{T}_{(\succsim_{AI}, u)\text{-}r} = \mathcal{T}_{(\succsim_{AB}, u)\text{-}r}. \quad (10)$$

Carbonell-Nicolau and Llavador (2021b) established the equivalence in (9) specifically for the relative Lorenz and the relative Foster-Wolfson pre-orders. However, as demonstrated by Theorem 3, this relationship extends more broadly. In fact, the equivalence holds for any inequality or bipolarization measure consistent with these pre-orders, and it also carries over to the class of absolute pre-orders.

Proof of Theorem 3. We first show that

$$\mathcal{T}_{(\succ_{RI}, u) \cdot r} = \mathcal{T}_{(\succ'_{RL}, u) \cdot r}. \quad (11)$$

Any relative inequality pre-order $\succ_{RI}$ is consistent with $\succ'_{RL}$. Consequently, $\mathcal{T}_{(\succ_{RI}, u) \cdot r} \supseteq \mathcal{T}_{(\succ'_{RL}, u) \cdot r}$. The containment $\mathcal{T}_{(\succ_{RI}, u) \cdot r} \subseteq \mathcal{T}_{(\succ'_{RL}, u) \cdot r}$ follows from [Proposition 1](#) and [Theorem 1](#). This establishes (11).

Next, we show that

$$\mathcal{T}_{(\succ_{RB}, u) \cdot r} = \mathcal{T}_{(\succ_{RFW}, u) \cdot r}, \quad (12)$$

where $\succ_{RFW}$ represents the relative Foster-Wolfson pre-order (see [Foster and Wolfson, 2010](#); [Wang and Tsui, 2000](#); [Chakravarty, 2009, 2015](#)).¹² Any relative bipolarization pre-order $\succ_{RB}$ is consistent with $\succ_{RFW}$ (i.e., $\succ_{RFW} \subseteq \succ_{RB}$) (see, e.g., [Chakravarty, 2009](#), Theorem 4.3, p. 119). Consequently, $\mathcal{T}_{(\succ_{RB}, u) \cdot r} \supseteq \mathcal{T}_{(\succ_{RFW}, u) \cdot r}$. The containment $\mathcal{T}_{(\succ_{RB}, u) \cdot r} \subseteq \mathcal{T}_{(\succ_{RFW}, u) \cdot r}$ follows from [Proposition 2](#) and [Theorem 1](#). This establishes (12).

The equivalence in (9) is a consequence of (11), (12), and the equality $\mathcal{T}_{(\succ'_{RL}, u) \cdot r} = \mathcal{T}_{(\succ_{RFW}, u) \cdot r}$, which is proven in [Carbonell-Nicolau and Llavador \(2021b\)](#), Theorem 4).

Next, we prove (10). We first show that

$$\mathcal{T}_{(\succ_{AI}, u) \cdot r} = \mathcal{T}_{(\succ'_{AL}, u) \cdot r}. \quad (15)$$

Note that [Proposition 1](#) and [Theorem 2](#) imply $\mathcal{T}_{(\succ_{AI}, u) \cdot r} \subseteq \mathcal{T}_{(\succ'_{AL}, u) \cdot r}$. The containment $\mathcal{T}_{(\succ_{AI}, u) \cdot r} \supseteq \mathcal{T}_{(\succ'_{AL}, u) \cdot r}$ follows from [Lemma 4](#) in [Appendix D](#). This establishes (15).

Next, we show that

$$\mathcal{T}_{(\succ_{AB}, u) \cdot r} = \mathcal{T}_{(\succ_{AFW}, u) \cdot r}, \quad (16)$$

where $\succ_{AFW}$ represents the absolute Foster-Wolfson pre-order ([Chakravarty, 2015](#), p. 48).¹³ Note that [Proposition 2](#) and [Theorem 2](#) imply $\mathcal{T}_{(\succ_{AB}, u) \cdot r} \subseteq \mathcal{T}_{(\succ_{AFW}, u) \cdot r}$. The containment $\mathcal{T}_{(\succ_{AB}, u) \cdot r} \supseteq \mathcal{T}_{(\succ_{AFW}, u) \cdot r}$ follows from [Lemma 5](#) in [Appendix D](#). This establishes (16).

Finally, [Lemma 6](#) in [Appendix D](#) yields

$$\mathcal{T}_{(\succ'_{AL}, u) \cdot r} = \mathcal{T}_{(\succ_{AFW}, u) \cdot r}. \quad (17)$$

The equivalence in (10) follows from (15), (16), and (17). ■

Secondly, polarization-reducing tax schedules decrease both inequality and bipolarization, whether measured in relative or absolute terms.

¹²The relative Foster-Wolfson pre-order ranks an income distribution $\mathbf{z}$ as more bipolarized than $\mathbf{z}'$ if the relative bipolarization curve of $\mathbf{z}$ does not lie below that of $\mathbf{z}'$. The relative bipolarization curve (RB) is constructed as follows. Given $\alpha \in [0, 1]$ and any income distribution $\mathbf{z} \in \mathcal{Z}_n$, define

$$RB(\mathbf{z}, \alpha) := \begin{cases} \frac{1}{nz_m} \sum_{j \leq i < m} (z_m - z_i) & \text{if } \alpha = \frac{j}{n} \text{ for some } j \in \{1, \dots, \frac{n}{2}\}, \\ \frac{1}{nz_m} \sum_{m \leq i \leq j} (z_i - z_m) & \text{if } \alpha = \frac{j}{n} \text{ for some } j \in \{\frac{n}{2} + 1, \dots, n\}, \\ 1 & \text{if } \alpha = 0, \end{cases} \quad (13)$$

and

$$RB(\mathbf{z}, \alpha) := \lambda RB\left(\mathbf{z}, \frac{j}{n}\right) + (1 - \lambda) RB\left(\mathbf{z}, \frac{j+1}{n}\right), \quad \text{if } \alpha = \lambda \left(\frac{j}{n}\right) + (1 - \lambda) \left(\frac{j+1}{n}\right), \quad (14)$$

where $\lambda \in (0, 1)$ and $j \in \{0, 1, \dots, n-1\}$.

¹³The absolute Foster-Wolfson pre-order results from comparing the absolute bipolarization curves, which are obtained by scaling up the relative bipolarization curves by their median. See [footnote 12](#).

Theorem 4. For any $u \in \mathcal{U}$ and any two relative inequality and bipolarization pre-orders, $\succ_{RI}$ and $\succ_{RB}$,

$$\mathcal{T}_{(\succ_{RP}, u) \cdot r} \subseteq \mathcal{T}_{(\succ_{RI}, u) \cdot r} = \mathcal{T}_{(\succ_{RB}, u) \cdot r}.^{14} \quad (18)$$

For any $u \in \mathcal{U}$ and any two absolute inequality and bipolarization pre-orders, $\succ_{AI}$ and $\succ_{AB}$,

$$\mathcal{T}_{(\succ_{AP}, u) \cdot r} \subseteq \mathcal{T}_{(\succ_{AI}, u) \cdot r} = \mathcal{T}_{(\succ_{AB}, u) \cdot r}. \quad (19)$$

Proof. The equality in (18) follows from Theorem 3. To see that $\mathcal{T}_{(\succ_{RP}, u) \cdot r} \subseteq \mathcal{T}_{(\succ_{RI}, u) \cdot r}$, note that because $\succ_{RP}$ is scale-invariant and satisfies BSM (Proposition 3), Theorem 1 implies $\mathcal{T}_{(\succ_{RP}, u) \cdot r} \subseteq \mathcal{T}_{(\succ'_{RL}, u) \cdot r}$. Now combine this inclusion with (11). The proof of (19) is analogous. ■

Finally, we show that the reverse inclusions

$$\mathcal{T}_{(\succ_{RP}, u) \cdot r} \supseteq \mathcal{T}_{(\succ_{RI}, u) \cdot r} \quad \text{and} \quad \mathcal{T}_{(\succ_{AP}, u) \cdot r} \supseteq \mathcal{T}_{(\succ_{AI}, u) \cdot r}$$

are not true in general. This means that there exist tax schedules that are inequality and bipolarization-reducing but not polarization-reducing, as illustrated in the following example and further explained in the discussion following Theorem 5.

Example 1. Consider the Cobb-Douglas utility function $u(c, l) = c(1 - l)$ and the progressive two-bracket tax schedule

$$T(y) = \begin{cases} -0.5 & \text{if } 0 \leq y \leq 0.1, \\ -0.55 + 0.5y & \text{if } y > 0.1. \end{cases}$$

We know that, for the Cobb-Douglas family of utility functions, $\mathcal{T}_{(\succ'_{RL}, u) \cdot r} = \mathcal{T}_{m\text{-}prog}$ (Remark 3 in Carbonell-Nicolau and Llavador, 2018). By (11), $\mathcal{T}_{(\succ_{RI}, u) \cdot r} = \mathcal{T}_{(\succ'_{RL}, u) \cdot r}$. Hence $T \in \mathcal{T}_{(\succ_{RI}, u) \cdot r}$. Choose $\mathbf{a} = (0.4, 0.5, 6)$, and compute the income distributions

$$\mathbf{y}(\mathbf{a}, 0) = (0.2, 0.25, 3) \quad \text{and} \quad \mathbf{x}(\mathbf{a}, T) = (0.5, 0.5, 3.25).$$

The corresponding relative polarization index values (for $\alpha = 1.6$) are

$$P_R(\mathbf{y}(\mathbf{a}, 0)) = 10.8K < 15.1K = P_R(\mathbf{x}(\mathbf{a}, T)).^{15}$$

Consequently, $T \notin \mathcal{T}_{(\succ_{RP}, u) \cdot r}$.

In the absolute case, we have $\mathcal{T}_{(\succ_{AI}, u) \cdot r} = \mathcal{T}_{m\text{-}prog}.$ ¹⁶ On the other hand, for $\alpha = 1.6$,

$$P_A(\mathbf{y}(\mathbf{a}, 0)) = 11.2K < 22.2K = P_A(\mathbf{x}(\mathbf{a}, T)).$$

Therefore, $T \in \mathcal{T}_{(\succ_{AI}, u) \cdot r}$ but $T \notin \mathcal{T}_{(\succ_{AP}, u) \cdot r}$.

While tax schedules designed to reduce inequality may not necessarily decrease polarization (as illustrated in Example 1), within the subclass of proportional tax schedules, the three social criteria—inequality, bipolarization, and polarization—are equivalent.

¹⁴The polarization orders $\succ_{RP}$ and $\succ_{AP}$, and hence the sets $\mathcal{T}_{(\succ_{RP}, u) \cdot r}$ and $\mathcal{T}_{(\succ_{AP}, u) \cdot r}$, depend on the parameter $\alpha \in (0, 1.6]$ of the Esteban–Ray index. To simplify notation, we suppress this dependence throughout, treating α as fixed but arbitrary within its admissible range. All results involving these sets hold for any such α .

¹⁵The use of $\alpha = 1.6$ is illustrative; in this example, the qualitative finding is robust across the valid parameter range.

¹⁶This holds because the Cobb-Douglas utility functions satisfy the conditions in Theorem 8.

Theorem 5. For any $u \in \mathcal{U}$ and any two relative inequality and bipolarization pre-orders, $\succ_{RI}$ and $\succ_{RB}$,

$$\mathcal{T}_{(\succ_{RP}, u)\text{-}r} \cap \mathcal{T}_{prop} = \mathcal{T}_{(\succ_{RI}, u)\text{-}r} \cap \mathcal{T}_{prop} = \mathcal{T}_{(\succ_{RB}, u)\text{-}r} \cap \mathcal{T}_{prop}.$$

For any $u \in \mathcal{U}$ and any two absolute inequality and bipolarization pre-orders, $\succ_{AI}$ and $\succ_{AB}$,

$$\mathcal{T}_{(\succ_{AP}, u)\text{-}r} \cap \mathcal{T}_{prop} = \mathcal{T}_{(\succ_{AI}, u)\text{-}r} \cap \mathcal{T}_{prop} = \mathcal{T}_{(\succ_{AB}, u)\text{-}r} \cap \mathcal{T}_{prop}.$$

Proof. By [Theorem 4](#), it suffices to show that

$$\mathcal{T}_{(\succ_{RP}, u)\text{-}r} \supseteq \mathcal{T}_{(\succ_{RI}, u)\text{-}r} \cap \mathcal{T}_{prop} \quad \text{and} \quad \mathcal{T}_{(\succ_{AP}, u)\text{-}r} \supseteq \mathcal{T}_{(\succ_{AI}, u)\text{-}r} \cap \mathcal{T}_{prop}. \quad (20)$$

To see that the first containment holds, suppose that $T \in \mathcal{T}_{(\succ_{RI}, u)\text{-}r} \cap \mathcal{T}_{prop}$. Then, by [\(11\)](#), $T \in \mathcal{T}_{(\succ'_{RL}, u)\text{-}r} \cap \mathcal{T}_{prop}$. Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathcal{A}_n$. Because T is $(\succ'_{RL}, u)\text{-}r$, [Lemma 2](#) in [Carbonell-Nicolau and Llavador \(2021b\)](#) implies that

$$\frac{x^u(a_1, T)}{y^u(a_1, 0)} \geq \dots \geq \frac{x^u(a_n, T)}{y^u(a_n, 0)}.$$

Therefore, given i, j with $j \geq i$, we have

$$\frac{x^u(a_i, T)}{y^u(a_i, 0)} \geq \frac{x^u(a_j, T)}{y^u(a_j, 0)},$$

implying that

$$\ln y^u(a_j, 0) - \ln y^u(a_i, 0) \geq \ln x^u(a_j, T) - \ln x^u(a_i, T).$$

In addition, because $u \in \mathcal{U}$, $a_j > a_i$ implies

$$y^u(a_i, 0) < y^u(a_j, 0) \quad \text{and} \quad x^u(a_i, T) < x^u(a_j, T),$$

where the last inequality uses the fact that T is proportional. Consequently, letting

$$\mathbf{y} = (y^u(a_1, 0), \dots, y^u(a_n, 0)) \quad \text{and} \quad \mathbf{x} = (x^u(a_1, T), \dots, x^u(a_n, T)),$$

we have

$$\begin{aligned} P_R(\mathbf{y}) &= K \sum_{i=1}^n \sum_{j=1}^n \pi_{\mathbf{y}}(y^u(a_i, 0))^\alpha |\ln y^u(a_i, 0) - \ln y^u(a_j, 0)| \\ &\geq K \sum_{i=1}^n \sum_{j=1}^n \pi_{\mathbf{x}}(x^u(a_i, T))^\alpha |\ln x^u(a_i, T) - \ln x^u(a_j, T)| = P_R(\mathbf{x}), \end{aligned}$$

implying that

$$(y^u(a_1, 0), \dots, y^u(a_n, 0)) \succ_{RP} (x^u(a_1, T), \dots, x^u(a_n, T)).$$

Hence, $T \in \mathcal{T}_{(\succ_{RP}, u)\text{-}r}$. This establishes the first containment in [\(20\)](#).¹⁷

¹⁷ As shown in [Appendix C](#), the reverse inclusion is not true in general:

$$\mathcal{T}_{(\succ_{RP}, u)\text{-}r} \not\subseteq \mathcal{T}_{(\succ_{RI}, u)\text{-}r} \cap \mathcal{T}_{prop}.$$

On the other hand, if one requires that tax schedules be $(\succ_{RP}, u)\text{-}r$ for *any*—rather than for a fixed—population size n , then the inclusion holds: $\mathcal{T}_{(\succ_{RP}, u)\text{-}r} \subseteq \mathcal{T}_{(\succ_{RI}, u)\text{-}r} \cap \mathcal{T}_{prop}$. In this case, a tax schedule is polarization-reducing if and only if it is proportional and inequality-reducing.

To see that the second containment in (20) holds, suppose that $T \in \mathcal{T}_{(\succ_{AL}, u)\text{-}r} \cap \mathcal{T}_{prop}$. Then (15) implies $T \in \mathcal{T}_{(\succ'_{AL}, u)\text{-}r} \cap \mathcal{T}_{prop}$. Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathcal{A}_n$. Because T is $(\succ'_{AL}, u)\text{-}r$, Lemma 1 implies that for i, j with $j \geq i$, we have

$$y^u(a_j, 0) - y^u(a_i, 0) \geq x^u(a_j, T) - x^u(a_i, T).$$

In addition, because $u \in \mathcal{U}$, $a_j > a_i$ implies

$$y^u(a_i, 0) < y^u(a_j, 0) \quad \text{and} \quad x^u(a_i, T) < x^u(a_j, T),$$

where the last inequality uses the fact that T is proportional. Consequently, letting

$$\mathbf{y} = (y^u(a_1, 0), \dots, y^u(a_n, 0)) \quad \text{and} \quad \mathbf{x} = (x^u(a_1, T), \dots, x^u(a_n, T)),$$

we have

$$\begin{aligned} P_A(\mathbf{y}) &= K \sum_{i=1}^n \sum_{j=1}^n \pi_{\mathbf{y}}(y^u(a_i, 0))^\alpha |y^u(a_i, 0) - y^u(a_j, 0)| \\ &\geq K \sum_{i=1}^n \sum_{j=1}^n \pi_{\mathbf{x}}(x^u(a_i, T))^\alpha |x^u(a_i, T) - x^u(a_j, T)| = P_A(\mathbf{x}), \end{aligned}$$

implying that

$$(y^u(a_1, 0), \dots, y^u(a_n, 0)) \succ_{AP} (x^u(a_1, T), \dots, x^u(a_n, T)).$$

Hence, $T \in \mathcal{T}_{(\succ_{AP}, u)\text{-}r}$.¹⁸ ■

We now examine why the equivalences in Theorem 5 break down beyond proportional tax schedules. The reason resides in the bunching around the income levels that separate tax brackets. In a Mirrlees framework, the tax brackets create ‘kinks’ in the budget set where many individuals with different underlying skills bunch into the same post-tax income. Bunching reduces the distance between individuals, lowering inequality and alienation, and hence bipolarization. On the other hand, by forcing individuals into a dense, homogeneous income cluster, the progressive tax increases identification, which may actually increase polarization despite making society more equal. Proportional taxation reduces alienation and avoids the bunching that reinforces identification, thereby mitigating all three phenomena.

Consider the distributions in Figure 1: distribution $\mathbf{y}$ is completely tripolarized, while distribution $\mathbf{x}$ is completely bipolarized. When comparing these distributions, the polarization assessment becomes complex and multifaceted. In $\mathbf{y}$, the increased distance between the first two poles and the third pole accentuates the sense of alienation, suggesting a higher degree of polarization. Conversely, in $\mathbf{x}$, while the poles are closer and alienation appears reduced, the merger of the two left poles from $\mathbf{y}$ intensifies group identification, which also represents a form of polarization. This comparison reveals a fundamental ambiguity: in some cases, polarization cannot be singularly defined or measured.

¹⁸As in the relative polarization case (see footnote 17), the reverse inclusion is not true in general:

$$\mathcal{T}_{(\succ_{AP}, u)\text{-}r} \not\subseteq \mathcal{T}_{(\succ_{AL}, u)\text{-}r} \cap \mathcal{T}_{prop}.$$

Consider the linear tax in Example 1 and $n = 2$. For any $a_1 < a_2$, $(y(a_2, 0) - y(a_1, 0)) \geq (x(a_2, T) - x(a_1, T))$ and hence the non-proportional tax $T \in \mathcal{T}_{(\succ_{AP}, u)\text{-}r}$, since $P_A(a, a') = K|a - a'|$.

While $\mathcal{T}_{(\succ_{AP}, u)\text{-}r} \not\subseteq \mathcal{T}_{prop}$, as we have illustrated, if one requires that tax schedules be $(\succ_{AP}, u)\text{-}r$ for any—rather than for a fixed—population size n , then the inclusion holds: $\mathcal{T}_{(\succ_{AP}, u)\text{-}r} \subseteq \mathcal{T}_{prop}$. In this case, a tax schedule is polarization-reducing if and only if it is proportional.

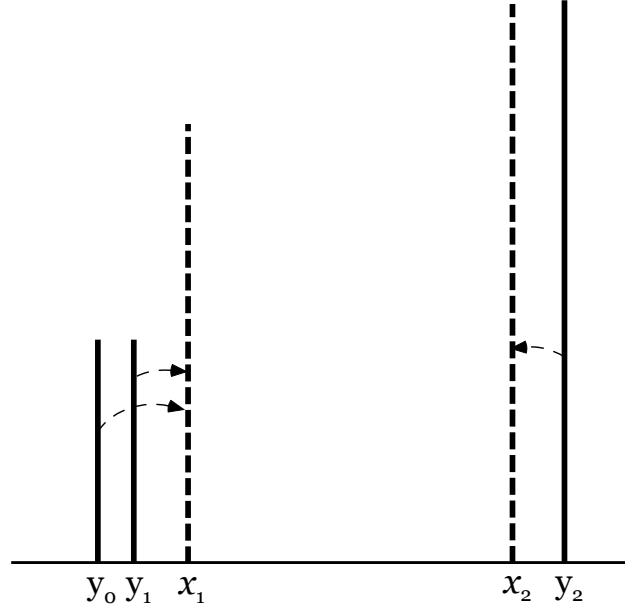


Figure 1: Distribution $\mathbf{y} = (y_0, y_1, y_2)$ shows higher inequality and bipolarity than $\mathbf{x} = (x_1, x_2)$. However, $\mathbf{y}$ may present higher or lower polarization than $\mathbf{x}$ depending on the share of the population on each income level and the distances between incomes.

The question then arises: How do measures of polarization handle this ambiguity? The Foster-Wolfson measure of polarization, discussed previously, unequivocally ranks distribution $\mathbf{y}$ as more bipolarized due to its adherence to the Increased Spread (IS) axiom. The Esteban–Ray index must also resolve the ambiguity, since it induces a complete ranking over distributions. However, unlike the Foster-Wolfson measure, this index does not consistently rank distributions like $\mathbf{y}$ as more polarized than $\mathbf{x}$; its assessment varies depending on the specific characteristics of the distributions being compared (see section 3.4 in [Esteban and Ray, 1994](#)).

Having explored the tension between bipolarization and the Esteban–Ray index, we can now better understand the failure of non-proportional tax schedules to reduce polarization, a failure that stems from cases where the Esteban–Ray index rates distributions like $\mathbf{x}$ from [Figure 1](#) as more polarized than $\mathbf{y}$.

Consider first the two-bracket tax schedule from [Example 1](#). For $\mathbf{a} = (0.4, 0.5, 6)$, the tax-free and after-tax distributions are

$$\mathbf{y}(\mathbf{a}, 0) = (0.2, 0.25, 3) \quad \text{and} \quad \mathbf{x}(\mathbf{a}, T) = (0.5, 0.5, 3.25),$$

respectively.

Note that the tax schedule consolidates the income of the two poorest individuals at the same after-tax level of \$0.5. This phenomenon, known as “bunching,” occurs at the kink point—the threshold between the two tax brackets—where the marginal tax rate changes. The convex structure of the tax function T creates this clustering effect, effectively compressing the lower end of the income distribution.

To make these two distributions directly comparable and draw the analogy with the example in [Figure 1](#), we can scale $\mathbf{y}(\mathbf{a}, 0)$ so that the transformation of $\mathbf{y}(\mathbf{a}, 0)$ has the same total income as $\mathbf{x}(\mathbf{a}, T)$:

$$\beta \mathbf{y}(\mathbf{a}, 0) = \beta(0.2, 0.25, 3) \approx (0.25, 0.31, 3.69),$$

where

$$\beta = \frac{0.5 + 0.5 + 3.25}{0.2 + 0.25 + 3} \approx 1.23.$$

Since

$$\beta \mathbf{y}(\mathbf{a}, 0) \sim_{RP} \mathbf{y}(\mathbf{a}, 0) \quad \text{and} \quad P_R(\mathbf{y}(\mathbf{a}, 0)) < P_R(\mathbf{x}(\mathbf{a}, T)),$$

the Esteban–Ray index resolves the identification-alienation trade-off by emphasizing the tax-induced pole consolidation. The tax merges two left poles into a single cluster, increasing identification, which the index deems more significant than the reduced overall distribution spread.¹⁹

3.3.2 Marginal-Rate Progressivity as a Necessary Condition

In this subsection, we demonstrate that any spread-reducing tax schedule must feature marginal-rate progressivity. This necessary condition holds true irrespective of the specific social pre-order employed to evaluate income distributions.

Theorem 6. *For any $u \in \mathcal{U}$ and any social pre-order $\succsim$ satisfying **BSM** and either **SI** or **TI**,*

$$\mathcal{T}_{(\succsim, u)\text{-}r} \subseteq \mathcal{T}_{m\text{-}prog}.$$

The significance of this result lies in its universal applicability across diverse social normative criteria, encompassing all relative and absolute measures of inequality and polarization.

Proof of Theorem 6. First, from Theorem 3 in Carbonell-Nicolau and Llavador (2021b), $\mathcal{T}_{(\succsim'_{RL}, u)\text{-}r} \subseteq \mathcal{T}_{m\text{-}prog}$. It follows from Theorem 1 that if $\succsim$ satisfies **SI** then $\mathcal{T}_{(\succsim, u)\text{-}r} \subseteq \mathcal{T}_{m\text{-}prog}$. Next, let $\succsim$ satisfy **TI**. By Theorem 2, it is sufficient to show that if $T \notin \mathcal{T}_{m\text{-}prog}$ then $T \notin \mathcal{T}_{(\succsim'_{AL}, u)\text{-}r}$. Let y^u and x^u be any two pre-tax and post-tax income functions. The map $a \mapsto y^u(a, 0)$ is continuous on $\mathbb{R}_{++}$, while the map $a \mapsto x^u(a, T)$ is non-decreasing (Lemma 1 in Carbonell-Nicolau and Llavador, 2018) and has at least one discontinuity point (Lemma 4 in Carbonell-Nicolau and Llavador, 2018). Let $a^* \in \mathbb{R}_{++}$ be one such discontinuity point. Then

$$\lim_{a \uparrow a^*} x^u(a, T) < \lim_{a \downarrow a^*} x^u(a, T) \quad \text{and} \quad \lim_{a \uparrow a^*} y^u(a, 0) = \lim_{a \downarrow a^*} y^u(a, 0),$$

implying that

$$\lim_{a \uparrow a^*} x^u(a, T) - \lim_{a \downarrow a^*} x^u(a, T) > \lim_{a \uparrow a^*} y^u(a, 0) - \lim_{a \downarrow a^*} y^u(a, 0) = 0.$$

Hence, we can find $a' > a^* > 0$ violating condition (7) in Lemma 1, and so $T \notin \mathcal{T}_{(\succsim'_{AL}, u)\text{-}r}$, as we wanted to prove. $\blacksquare$

3.3.3 Full Characterizations

Building on the preceding analysis, we can now derive comprehensive characterizations of spread-reducing tax schedules for each of the social pre-orders previously examined: inequality, bipolarization, and polarization pre-orders.

For relative inequality pre-orders, the results in this paper can be used to extend the existing characterizations of inequality and bipolarization-reducing tax schedules for the specific relative Lorenz and Foster-Wolfson pre-orders (Carbonell-Nicolau and Llavador, 2018, 2021a,b) to any extensions of these pre-orders.

¹⁹A similar example can be constructed for the absolute polarization case.

A full characterization of inequality and bipolarization-reducing tax schedules with respect to any relative inequality and bipolarization pre-order requires a condition on the income elasticity with respect to ability. For $(a, b) \in \mathbb{R}_{++} \times \mathbb{R}_+$, let

$$\zeta^u(a, b) := \frac{\partial(al^u(a, b) + b)}{\partial a} \cdot \frac{a}{al^u(a, b) + b}$$

be the elasticity of income with respect to ability at (a, b) .

For $b \geq 0$ and $R \subseteq [0, 1]$, let $\mathcal{U}_\zeta(b, R)$ be the set of all $u \in \mathcal{U}$ such that

$$\zeta^u((1-r)a, b') \leq \zeta^u(a, 0), \quad \text{for all } (a, b', r) \in \mathbb{R}_{++} \times [b, \infty) \times R. \quad (21)$$

Theorem 7. For $u \in \mathcal{U}$, $b \geq 0$, $R \subseteq [0, 1]$, and any two relative inequality and bipolarization pre-orders $\succ_{RI}$ and $\succ_{RB}$,

$$\mathcal{T}_{(\succ_{RI}, u)-r} = \mathcal{T}_{(\succ_{RB}, u)-r} \subseteq \mathcal{T}_{m-prog}$$

and

$$\mathcal{T}_{m-prog}(b, R) \subseteq \mathcal{T}_{(\succ_{RI}, u)-r} = \mathcal{T}_{(\succ_{RB}, u)-r} \Leftrightarrow u \in \mathcal{U}_\zeta(b, R). \quad (22)$$

Proof. Proposition 1, Theorem 3, and Theorem 6 give the first containment. The equivalence follows from Theorem 3 in Carbonell-Nicolau and Llavador (2021b), Theorem 3, and the fact that $\mathcal{T}_{(\succ_{RI}, u)-r} = \mathcal{T}_{(\succ'_{RL}, u)-r}$ (see (11)). ■

The equivalence in (22) states that the members of the set $\mathcal{T}_{m-prog}(b, R)$ of all marginal-rate progressive tax schedules in $\mathcal{T}_{m-prog}$ whose intercept is greater than or equal to b and whose marginal tax rates lie in R are all inequality-reducing if and only if $u \in \mathcal{U}_\zeta(b, R)$, i.e., if and only if the elasticity of income with respect to ability satisfies condition (21).

Theorem 7 was initially established for the specific case of the relative Lorenz pre-order and the relative Foster-Wolfson pre-order by Carbonell-Nicolau and Llavador (2021a,b). However, our current analysis demonstrates that the applicability of this theorem extends beyond its original context. In fact, it holds universally across the entire class of relative inequality pre-orders—which includes all possible extensions of the relative Lorenz pre-order—as well as the class of relative bipolarization pre-orders, which encompasses all extensions of the relative Foster-Wolfson pre-order.

In the special case when the lower bound on the intercept of a tax schedule is zero and the marginal tax rates can take values anywhere in the interval $[0, 1]$, Theorem 7 immediately gives the following result, which was first proven for the particular case of the relative Lorenz criterion in Carbonell-Nicolau and Llavador (2018, Corollary 3).

Corollary 1 (to Theorem 7). For $u \in \mathcal{U}$ and any two relative inequality and bipolarization pre-orders $\succ_{RI}$ and $\succ_{RB}$,

$$\mathcal{T}_{(\succ_{RI}, u)-r} = \mathcal{T}_{(\succ_{RB}, u)-r} = \mathcal{T}_{m-prog} \Leftrightarrow u \in \mathcal{U}_\zeta(0, [0, 1]).$$

The reader is referred to Carbonell-Nicolau and Llavador (2021a) for applications of Theorem 7 to the families of constant elasticity of substitution (CES) and quasilinear utility functions.

Our analysis yields a novel result that parallels Theorem 7, extending its application to absolute inequality and bipolarization pre-orders. In the absolute context, the class of

²⁰Refer to Carbonell-Nicolau and Llavador (2021a) for an interpretation of condition (21) based on a decomposition of the inequality in (21) into two conditions on the wage elasticity of income, each capturing different aspects of the transition between before-tax and after-tax income distributions.

preferences that fully characterizes inequality and bipolarization-reducing tax schedules diverges from the previously established $\mathcal{U}_\zeta(b, R)$. This distinction requires the introduction of a new family of preferences.

For $(a, b) \in \mathbb{R}_{++} \times \mathbb{R}_+$, let

$$\xi^u(a, b) := \frac{\partial l^u(a, b)}{\partial a} \cdot \frac{a}{l^u(a, b)}$$

be the *wage elasticity of the labor supply* at (a, b) . For $b \geq 0$ and $R \subseteq [0, 1]$, let $\mathcal{U}_\xi(b, R)$ be the set of all $u \in \mathcal{U}$ such that

$$\begin{aligned} y^u((1-r)a, b')[1 + \xi^u((1-r)a, b')] \\ \leq y^u(a, 0)[1 + \xi^u(a, 0)], \quad \text{for all } (a, b', r) \in \mathbb{R}_{++} \times [b, \infty) \times R, \end{aligned} \quad (23)$$

Observe that, for any $b \geq 0$ and $R \subseteq [0, 1]$, the set $\mathcal{U}_\xi(b, R)$ is nonempty as, for example, any utility function of the Cobb-Douglas form $u(c, l) := Ac^\alpha(1-l)^{1-\alpha}$, where $A > 0$ and $\alpha \in (0, 1)$, satisfies (23).

The preference classes $\mathcal{U}_\zeta(b, R)$ and $\mathcal{U}_\xi(b, R)$ are connected through a direct relationship between their defining elasticities. The wage elasticity of income (ζ^u) and the wage elasticity of labor supply (ξ^u) are linked by the following identity:

$$\zeta^u(a, b) = \frac{y^u(a, b)}{y^u(a, b) + b} (1 + \xi^u(a, b)).$$

Using this relationship, we can reformulate the identifying condition for $\mathcal{U}_\zeta(b, R)$ given in (21) in terms of ξ^u :

$$\begin{aligned} y^u((1-r)a, b')[1 + \xi^u((1-r)a, b')] \\ \leq [y^u((1-r)a, b') + b'][1 + \xi^u(a, 0)], \quad \text{for all } (a, b', r) \in \mathbb{R}_{++} \times [b, \infty) \times R. \end{aligned} \quad (24)$$

Comparing (23) with (24), we find that their relative stringency depends entirely on the relationship between two variables:

- The net income in a tax-free environment, $y^u(a, 0)$.
- The net income under a subsidy b' and proportional tax rate r , $y^u((1-r)a, b') + b'$.

The relationship between $y^u(a, 0)$ and $y^u((1-r)a, b') + b'$ depends dynamically on three parameters: the wage rate a , the tax rate r , and the subsidy level b' . The subsidy increases non-labor income, while the tax reduces the net wage, potentially curtailing individual work hours. These opposing forces create ambiguous effects on net income, which may exceed or fall below the tax-free income level, depending on the relative magnitudes of the income and substitution effects induced by the tax rate and subsidy. Since $y^u(a, 0)$ and $y^u((1-r)a, b') + b'$ fluctuate as functions of these parameters, neither condition (23) nor (24) implies the other. Consequently, the preference classes $\mathcal{U}_\zeta(b, R)$ and $\mathcal{U}_\xi(b, R)$ are non-nested, with their relative containment contingent on the specific economic environment.²¹

The following is the analogue of **Theorem 7** in the absolute context.

²¹For example, for the Cobb-Douglas function $u(c, l) = c(1-l)$, we have

$$y^u(a, 0) = 2/a \quad \text{and} \quad y^u((1-r)a, b) + b = 2/((1-r)a + b).$$

For $b = 1$ and $r = 0.1$,

$$2/a > 2/((1-r)a + b) \quad \text{if } a < 10,$$

$$2/a < 2/((1-r)a + b) \quad \text{if } a > 10.$$

Theorem 8. For $u \in \mathcal{U}$, $b \geq 0$, $R \subseteq [0, 1)$, and any two absolute inequality and bipolarization pre-orders $\succ_{AI}$ and $\succ_{AB}$,

$$\mathcal{T}_{(\succ_{AI}, u)-r} = \mathcal{T}_{(\succ_{AB}, u)-r} \subseteq \mathcal{T}_{m-prog} \quad (25)$$

and

$$\mathcal{T}_{m-prog}(b, R) \subseteq \mathcal{T}_{(\succ_{AI}, u)-r} = \mathcal{T}_{(\succ_{AB}, u)-r} \Leftrightarrow u \in \mathcal{U}_\xi(b, R). \quad (26)$$

The equivalence in (26) states that the members of the set $\mathcal{T}_{m-prog}(b, R)$ of all marginal-rate progressive tax schedules in $\mathcal{T}_{m-prog}$ whose intercept is greater than or equal to b and whose marginal tax rates lie in R are all inequality-reducing in the absolute sense if and only if $u \in \mathcal{U}_\xi(b, R)$, i.e., if and only if the wage elasticity of labor satisfies condition (23).

Remark 1. Theorem 8 contrasts sharply with Moyes (1988), who shows that under exogenous income, tax liabilities being monotonically increasing in income is necessary and sufficient for universal absolute inequality reduction—a condition so mild it admits marginal-rate regressive schedules. The endogenous framework tightens this considerably: only a subclass of marginal-rate progressive schedules qualifies.

The families $\mathcal{U}_\xi$ and $\mathcal{U}_\zeta$ constrain behavioral responses to ensure inequality reduction by comparing a common measure of post-tax dispersion against distinct benchmarks. In both conditions (23) and (24), the left-hand side represents post-tax income slope responsiveness. The divergence lies solely in the right-hand side: $\mathcal{U}_\xi$ scales the benchmark by pre-tax earned income, while $\mathcal{U}_\zeta$ scales it by post-tax total resources. This distinction dictates condition stringency under different tax structures.

For lump-sum subsidies, mechanical effects relax the relative condition but tighten the absolute one. The subsidy b' mechanically inflates the post-tax total resources in the relative benchmark (24), expanding the right-hand side independent of behavioral responses. Conversely, the absolute benchmark (23) relies on fixed pre-tax earned income, offering no mechanical buffer. Consequently, the absolute condition is stricter, placing the burden on behavioral responses to prevent gap widening. This aligns with the intuition that lump-sum subsidies mechanically curb relative dispersion *before* behavioral responses are factored in, while leaving absolute gaps unchanged.

Under proportional taxation, mechanical effects reverse this stringency. The tax reduces net wages, shrinking the relative benchmark and tightening that condition. Conversely, the absolute benchmark remains fixed at pre-tax earned income, providing a buffer against the compressed post-tax slope. Thus, the absolute condition is easier to satisfy because the tax mechanically compresses absolute gaps *before* behavioral responses without shrinking the benchmark, whereas the relative condition becomes stricter since proportional taxes fail to mechanically reduce relative ratios.²²

We use the following two lemmas, whose proofs are relegated to [Appendix E](#), to establish [Theorem 8](#). The two lemmas combined imply that it suffices to obtain conditions on preferences ensuring that the linear progressive tax schedules are inequality-reducing.

Lemma 2. For $u \in \mathcal{U}$, $b \geq 0$, and $R \subseteq [0, 1)$, $\mathcal{T}_{m-prog}(b, R) \subseteq \mathcal{T}_{(\succ'_{AL}, u)-r} \Leftrightarrow \mathcal{T}_{lin}(b, R) \subseteq \mathcal{T}_{(\succ'_{AL}, u)-r}$.

Lemma 3. For $u \in \mathcal{U}$, $b \geq 0$, and $R \subseteq [0, 1)$, $\mathcal{T}_{lin}(b, R) \subseteq \mathcal{T}_{(\succ'_{AL}, u)-r} \Leftrightarrow u \in \mathcal{U}_\xi(b, R)$.

Proof of Theorem 8. [Proposition 1](#), [Theorem 3](#), and [Theorem 6](#) give (25). The equivalence in (26) follows from [Lemma 2](#), [Lemma 3](#), [Theorem 3](#), and the fact that $\mathcal{T}_{(\succ_{AI}, u)-r} = \mathcal{T}_{(\succ'_{AL}, u)-r}$ (see (15)). ■

²²While the relative stringency of the elasticity conditions varies with the tax schedule, there are preferences (such as the Cobb-Douglas preferences) that satisfy both (23) and (24).

In the special case when the lower bound on the intercept of a tax schedule is zero and the marginal tax rates can take values anywhere in the interval $[0, 1)$, **Theorem 8** immediately gives the following equivalence.

Corollary 2 (to **Theorem 8**). *For any $u \in \mathcal{U}$ and any two absolute inequality and bipolarization pre-orders $\succ_{AI}$ and $\succ_{AB}$,*

$$\mathcal{T}_{(\succ_{AI}, u)\text{-}r} = \mathcal{T}_{(\succ_{AB}, u)\text{-}r} = \mathcal{T}_{m\text{-}prog} \Leftrightarrow u \in \mathcal{U}_\xi(0, [0, 1)).$$

Per the insights provided by **Theorem 5** and the subsequent discussion, we have established that only proportional tax schedules that reduce inequality and bipolarization consistently lead to a decrease in polarization as measured by the Esteban–Ray index. Such polarization-reducing tax schedules can be characterized as a corollary of **Theorem 5**, **Theorem 7**, and **Theorem 8**.

Theorem 9. *For any $u \in \mathcal{U}$, $b \geq 0$, $R \subseteq [0, 1)$, and any two relative inequality and bipolarization pre-orders $\succ_{RI}$ and $\succ_{RB}$,*

$$\mathcal{T}_{m\text{-}prog}(0, R) \subseteq \mathcal{T}_{(\succ_{RP}, u)\text{-}r} \cap \mathcal{T}_{prop} = \mathcal{T}_{(\succ_{RI}, u)\text{-}r} \cap \mathcal{T}_{prop} = \mathcal{T}_{(\succ_{RB}, u)\text{-}r} \cap \mathcal{T}_{prop} \Leftrightarrow u \in \mathcal{U}_\zeta(0, R).$$

For any $u \in \mathcal{U}$, $b \geq 0$, $R \subseteq [0, 1)$, and any two absolute inequality and bipolarization pre-orders $\succ_{AI}$ and $\succ_{AB}$,

$$\mathcal{T}_{m\text{-}prog}(0, R) \subseteq \mathcal{T}_{(\succ_{AP}, u)\text{-}r} \cap \mathcal{T}_{prop} = \mathcal{T}_{(\succ_{AI}, u)\text{-}r} \cap \mathcal{T}_{prop} = \mathcal{T}_{(\succ_{AB}, u)\text{-}r} \cap \mathcal{T}_{prop} \Leftrightarrow u \in \mathcal{U}_\xi(0, R).$$

4 Concluding remarks

This paper advances the theory of tax-mediated income distribution by introducing the Bimodal Spread Majorization (BSM) axiom—a unified framework that connects three distinct phenomena: income inequality, bipolarization, and polarization. The BSM axiom provides a rigorous foundation for analyzing how tax policies affect the spread of income across a population.

By embedding the BSM axiom within a social pre-order framework, we derive fundamental principles for designing tax policies that jointly address income inequality and polarization. This unified approach provides a theoretical foundation for progressive taxation and highlights its role in fostering equitable and socially cohesive societies.

The analysis yields three central results on the relation between income distribution and taxation.

First, inequality-reducing taxation is equivalent to bipolarization-reducing taxation, whereas polarization-reducing taxation is a more restrictive criterion. All polarization-reducing tax schedules are inequality-reducing, but the converse does not hold. Our equivalence result shows that, for a policymaker, reducing inequality and protecting the middle class (reducing bipolarization) are effectively the same objective. Any tax that compresses the overall spread also protects the center from hollowing out. On the other hand, the bunching effect at tax bracket thresholds creates a tension between inequality and polarization. While a progressive tax reduces the alienation between the rich and poor, it simultaneously increases identification. Proportional taxation, because it lacks the kinks of multi-bracket progressive systems, reduces the gaps between the rich and poor without new, dense social clusters.

Second, spread-reducing tax schedules must exhibit marginal-rate progressivity, characterized by non-decreasing marginal tax rates on income. Specifically, for any social pre-order satisfying BSM and either scale or translation invariance, order-reducing taxation must necessarily be marginal-rate progressive. Since all inequality, bipolarization, and polarization

measures are particular instances of social pre-orders satisfying these properties, marginal-rate progressivity is a necessary condition for any spread-reducing objective compatible with social cohesion across all three metrics.

Third, the paper identifies conditions under which progressive taxation can effectively increase equality and depolarization, emphasizing the importance of tax policy design in achieving these dual objectives.

As demonstrated in [Theorem 5](#), only proportional schedules ensure equivalence across inequality, bipolarization, and polarization reduction, avoiding the identification-enhancing bunching inherent in multi-bracket progressive systems. However, even within this proportional subclass, inequality reduction is not automatic; it depends on preference structures satisfying the elasticity conditions outlined in [Theorem 7](#) and [Theorem 8](#). The stringency of these conditions differs by normative criterion. As detailed in [Section 3.3.3](#), proportional taxation loosens the necessary and sufficient elasticity condition for absolute inequality reduction more than it does for relative inequality reduction. This is because proportional taxes mechanically compress absolute gaps—before behavioral responses are factored in—whereas they do not alter relative ratios. Thus, policymakers aiming to mitigate polarization via proportional taxes may find absolute equality goals more attainable than relative ones, contingent on labor supply responses.

Building on the established relationship between income tax progressivity, equality, and depolarization, a natural extension is to examine the connection between measures of tax progressivity and variance-sensitive normative criteria for evaluating income distributions. Specifically, does a higher degree of progressivity consistently lead to lower inequality and (bi)polarization? Promising insights are provided by [Le Breton et al. \(1996\)](#) for the specific case of exogenous income and the relative Lorenz pre-order. This extension constitutes a natural avenue for future research.

APPENDIX

A Equivalence to [Esteban and Ray \(1994\)](#)

We show that our formulation of relative polarization index in [\(3\)](#) is equivalent to the expression in [Esteban and Ray \(1994\)](#), which is

$$P_R(\mathbf{x}, \boldsymbol{\pi}) = K \sum_{i=1}^k \sum_{j=1}^k \pi_i^{1+\alpha} \pi_j |\ln x_i - \ln x_j|;$$

here, $\mathbf{x} = (x_1, \dots, x_k)$ represents the vector of distinct income levels in the distribution at hand, and $\boldsymbol{\pi} = (\pi_1, \dots, \pi_k)$ is a vector of densities, where each π_j ($j \in \{1, \dots, k\}$) represents the number of individuals with income level x_j ; the parameter K is positive and $\alpha \in (0, 1.6]$.

To see that the equivalence holds, consider an income distribution $\mathbf{z} = (z_1, \dots, z_n) \in \mathcal{Z}_n$. Its corresponding *density* is denoted by $(x_1, \dots, x_{k_z}; \pi_1, \dots, \pi_{k_z})$, i.e.,

$$\mathbf{z} = (\underbrace{x_1, \dots, x_1}_{\pi_1}, \underbrace{x_2, \dots, x_2}_{\pi_2}, \dots, \underbrace{x_{k_z}, \dots, x_{k_z}}_{\pi_{k_z}}),$$

where k_z represents the number of distinct income levels in the vector $\mathbf{z}$.

Letting $\Pi_j = \pi_1 + \dots + \pi_j$ (note that $\Pi_1 = \pi_1$), we can write

$$\begin{aligned}
P_R(\mathbf{z}) &= K \sum_{i=1}^n \sum_{j=1}^n \pi_z(z_i)^\alpha |\ln z_i - \ln z_j| \\
&= K \left(\sum_{i=1}^{\Pi_1} \pi_1^\alpha \sum_{j=1}^n |\ln x_1 - \ln z_j| + \sum_{i=\Pi_1+1}^{\Pi_2} \pi_2^\alpha \sum_{j=1}^n |\ln x_2 - \ln z_j| + \dots + \sum_{i=\Pi_{k_z-1}+1}^{\Pi_{k_z}} \pi_{k_z}^\alpha \sum_{j=1}^n |\ln x_{k_z} - \ln z_j| \right) \\
&= K \left(\pi_1^{\alpha+1} \sum_{j=1}^n |\ln x_1 - \ln z_j| + \pi_2^{\alpha+1} \sum_{j=1}^n |\ln x_2 - \ln z_j| + \dots + \pi_{k_z}^{\alpha+1} \sum_{j=1}^n |\ln x_{k_z} - \ln z_j| \right) \\
&= K \left(\pi_1^{\alpha+1} \sum_{j=1}^{k_z} \pi_j |\ln x_1 - \ln x_j| + \pi_2^{\alpha+1} \sum_{j=1}^{k_z} \pi_j |\ln x_2 - \ln x_j| + \dots + \pi_{k_z}^{\alpha+1} \sum_{j=1}^{k_z} \pi_j |\ln x_{k_z} - \ln x_j| \right) \\
&= K \sum_{i=1}^{k_z} \sum_{j=1}^{k_z} \pi_i^{\alpha+1} \pi_j |\ln x_i - \ln x_j| = P_R(\mathbf{x}, \boldsymbol{\pi}),
\end{aligned}$$

where we have used the fact that $\sum_{j=1}^n |\ln x_i - \ln z_j| = \sum_{j=1}^{k_z} \pi_j |\ln x_i - \ln x_j|$ for each i .

B Proof of Lemma 1

Lemma 1. For $u \in \mathcal{U}$, a tax schedule $T \in \mathcal{T}$ is $(\succ'_{AL}, u)$ -r if and only if, for any pre-tax and post-tax income functions y^u and x^u ,

$$y^u(a', 0) - y^u(a, 0) \geq x^u(a', T) - x^u(a, T), \quad \text{whenever } a' > a > 0. \quad (7)$$

Proof. [$\Leftarrow$] Fix $u \in \mathcal{U}$ and $T \in \mathcal{T}$ and suppose that (7) holds for any x^u and y^u . We must show that T is $(\succ'_{AL}, u)$ -r, i.e.,

$$(y^u(a_1, T), \dots, y^u(a_n, T)) \succ'_{AL} (x^u(a_1, 0), \dots, x^u(a_n, 0))$$

for every ability distribution $(a_1, \dots, a_n) \in \mathcal{A}_n$ and every pre-tax and post-tax income functions y^u and x^u .

Fix $(a_1, \dots, a_n) \in \mathcal{A}_n$, y^u , and x^u . By (7), we have

$$y^u(a_{i+1}, 0) - x^u(a_{i+1}, T) \geq y^u(a_i, 0) - x^u(a_i, T), \quad i \in \{1, \dots, n-1\}. \quad (27)$$

Let

$$\mathbf{y} = (y_1, \dots, y_n) := (y^u(a_1, 0), \dots, y^u(a_n, 0)),$$

$$\mathbf{x} = (x_1, \dots, x_n) := (x^u(a_1, T), \dots, x^u(a_n, T)),$$

and

$$\mu(\mathbf{y}) = \frac{\sum_{i=1}^n y_i}{n}, \quad \mu(\mathbf{x}) = \frac{\sum_{i=1}^n x_i}{n}.$$

Note that (27) can be rewritten as

$$\hat{y}_{i+1} - \hat{x}_{i+1} \geq \hat{y}_i - \hat{x}_i, \quad i \in \{1, \dots, n-1\}, \quad (28)$$

where

$$\hat{y}_i := y_i - \mu(\mathbf{y}) \quad \text{and} \quad \hat{x}_i := x_i - \mu(\mathbf{x}), \quad i \in \{1, \dots, n\}.$$

Since

$$\sum_{i=1}^n (\hat{x}_i - \hat{y}_i) = 0 \quad (29)$$

and (by (28)) $\hat{x}_i - \hat{y}_i$ is non-decreasing in i , there exists I such that

$$\begin{aligned}\hat{y}_i - \hat{x}_i &\leq 0, \quad i \in \{1, \dots, I\}, \\ \hat{y}_i - \hat{x}_i &\geq 0, \quad i \in \{I+1, \dots, n\}.\end{aligned}$$

Consequently,

$$\sum_{i=1}^k (\hat{x}_i - \hat{y}_i) \geq 0, \quad k \in \{1, \dots, I\}, \quad (30)$$

$$\sum_{i=k}^n (\hat{x}_i - \hat{y}_i) \leq 0, \quad k \in \{I+1, \dots, n\}. \quad (31)$$

Note that

$$\sum_{i=1}^k (\hat{x}_i - \hat{y}_i) = \sum_{i=1}^n (\hat{x}_i - \hat{y}_i) - \sum_{i=k+1}^n (\hat{x}_i - \hat{y}_i) \geq 0, \quad k \in \{I+1, \dots, n\}, \quad (32)$$

where the inequality follows from (29) and (31). Combining (30) and (32) yields

$$\sum_{i=1}^k (\hat{x}_i - \hat{y}_i) \geq 0, \quad k \in \{1, \dots, n\},$$

implying that $\mathbf{y} \succ'_{AL} \mathbf{x}$, as we sought.

[$\Rightarrow$] Suppose that there exist $a' > a > 0$, x^u , and y^u such that

$$y^u(a', 0) - y^u(a, 0) < x^u(a', T) - x^u(a, T). \quad (33)$$

It suffices to show that T is not $(\succ'_{AL}, u)$ -r, i.e.,

$$(y^u(a_1, 0), \dots, y^u(a_n, 0)) \not\succeq (x^u(a_1, T), \dots, x^u(a_n, T)) \quad (34)$$

for some $(a_1, \dots, a_n) \in \mathcal{A}_n$, y^u , and x^u .

Let $\mathbf{a} = (a_1, \dots, a_n)$ satisfy

$$a_1 = \dots = a_{n-1} = a < a' = a_n.$$

Then,

$$\begin{aligned}y^u(a_1, 0) - \frac{\sum_{i=1}^n y^u(a_i, 0)}{n} &= \frac{y^u(a, 0) - y^u(a', 0)}{n} \\ &> \frac{x^u(a, T) - x^u(a', T)}{n} = x^u(a_1, T) - \frac{\sum_{i=1}^n x^u(a_i, T)}{n}\end{aligned}$$

(where the inequality follows from (33)), implying that

$$y^u(a_1, 0) - \frac{\sum_{i=1}^n y^u(a_i, 0)}{n} > x^u(a_1, T) - \frac{\sum_{i=1}^n x^u(a_i, T)}{n},$$

which gives (34), as desired. ■

C Proof that $\mathcal{T}_{(\succ_{RP}, u)\text{-}r} \not\subseteq \mathcal{T}_{(\succ_{RI}, u)\text{-}r} \cap \mathcal{T}_{prop}$

We provide an example showing that the inclusion $\mathcal{T}_{(\succ_{RP}, u)\text{-}r} \subseteq \mathcal{T}_{(\succ_{RI}, u)\text{-}r} \cap \mathcal{T}_{prop}$ is not true in general.

Consider the case for $n = 2$. The relative polarization index becomes

$$P_R(z_1, z_2) = K |\ln(z_1) - \ln(z_2)|.$$

For any $\mathbf{a} = (a_1, a_2)$ with $a_1 \leq a_2$,

$$P_R(y(\mathbf{a}, 0)) \geq P_R(x(\mathbf{a}, T)) \Leftrightarrow \ln y(a_2, 0) - \ln y(a_1, 0) \geq \ln x(a_2, T) - \ln x(a_1, T),$$

which is equivalent to $\frac{x(a, T)}{y(a, 0)}$ being non-increasing in a . In combination with Lemma 2 in [Carbonell-Nicolau and Llavador \(2021b\)](#), this implies that, for $n = 2$,

$$\mathcal{T}_{(\succ_{RP}, u)\text{-}r} = \mathcal{T}_{(\succ_{RL}, u)\text{-}r}.$$

On the other hand, by Propositions 1 and 2 in [Carbonell-Nicolau and Llavador \(2021a\)](#), there exist non-linear, $(\succ'_{RL}, u)\text{-}r$ (hence $(\succ_{RP}, u)\text{-}r$) tax schedules for certain CES and quasi-linear utility functions. These cases would belong to $\mathcal{T}_{(\succ_{RP}, u)\text{-}r}$ but not to $\mathcal{T}_{(\succ_{RI}, u)\text{-}r} \cap \mathcal{T}_{prop}$.

D Consistency

Definition 9. A social pre-order $\succ$ is $\succ'$ -**consistent**, where $\succ'$ is another social pre-order, if the following two conditions are satisfied:

1. $\mathbf{x} \sim' \mathbf{y} \Rightarrow \mathbf{x} \sim \mathbf{y}$.
2. $\mathbf{x} \succ' \mathbf{y} \Rightarrow \mathbf{x} \succ \mathbf{y}$.

Lemma 4. An inequality pre-order satisfies **TI** if and only if it is $\succ'_{AL}$ -consistent.

Proof. Let $\succ$ be an inequality pre-order satisfying **TI**. If $\succ$ is $\succ'_{AL}$ -consistent, then it satisfies **TI** and **TP** because $\succ'_{AL}$ satisfies these axioms.

Conversely, suppose that $\succ$ satisfies **TI** and **TP**. Fix $\mathbf{x}$ and $\mathbf{y}$. Suppose that

$$\sum_i x_i \leq \sum_i y_i$$

(the case when $\sum_i x_i > \sum_i y_i$ can be handled similarly). Define ε by

$$n\varepsilon + \sum_i x_i = \sum_i y_i.$$

Note that

$$\mathbf{x}' \sim'_{AL} \mathbf{x},$$

where

$$\mathbf{x}' := (x_1 + \varepsilon, \dots, x_n + \varepsilon).$$

If $\mathbf{x} \sim'_{AL} \mathbf{y}$, then

$$\mathbf{x}' \sim'_{AL} \mathbf{x} \sim'_{AL} \mathbf{y},$$

implying $\mathbf{x}' \sim'_{AL} \mathbf{y}$. Hence, since $\mathbf{x}'$ and $\mathbf{y}$ have the same total income, we have $\mathbf{x}' = \mathbf{y}$, implying $\mathbf{x}' \sim \mathbf{y}$ (by reflexivity of $\succ$). Therefore, by **TI**, we have

$$\mathbf{x} \sim \mathbf{x}' \sim \mathbf{y},$$

implying $\mathbf{x} \sim \mathbf{y}$.²³

If $\mathbf{x} \succ'_{AL} \mathbf{y}$, then

$$\mathbf{x}' \sim'_{AL} \mathbf{x} \succ'_{AL} \mathbf{y},$$

implying $\mathbf{x}' \succ'_{AL} \mathbf{y}$. Consequently, because $\mathbf{x}'$ and $\mathbf{y}$ have the same income, Lemma B.1 in Marshall et al. (2011, chapter 1) implies that $\mathbf{x}'$ can be obtained from $\mathbf{y}$ by a finite sequence of progressive transfers, and so (by TP) $\mathbf{x}' \succ \mathbf{y}$. Therefore, by TI, we have

$$\mathbf{x} \sim \mathbf{x}' \succ \mathbf{y}.$$

Since $\succsim$ is reflexive and transitive, this yields $\mathbf{x} \succ \mathbf{y}$. ■

Lemma 5. *A bipolarization pre-order satisfies TI if and only if it is $\succsim_{AFW}$ -consistent.*

Proof. Let $\succsim$ be a bipolarization pre-order satisfying TI. If $\succsim$ is $\succsim_{AFW}$ -consistent, then it satisfies TI, IS, and IB because $\succsim_{AFW}$ satisfies these axioms.

Conversely, suppose that $\succsim$ satisfies TI, IS, and IB. Fix $\mathbf{x}$ and $\mathbf{y}$, with respective median incomes $m(\mathbf{x})$ and $m(\mathbf{y})$. Suppose that $m(\mathbf{x}) \leq m(\mathbf{y})$ (the case when $m(\mathbf{x}) > m(\mathbf{y})$ can be handled similarly). Let ε be defined by

$$m(\mathbf{x}) + \varepsilon = m(\mathbf{y}).$$

Then $m(\mathbf{x}') = m(\mathbf{y})$, where

$$\mathbf{x}' := (x_1 + \varepsilon, \dots, x_n + \varepsilon).$$

If $\mathbf{x} \sim_{AFW} \mathbf{y}$, then

$$\mathbf{x}' \sim_{AFW} \mathbf{x} \sim_{AFW} \mathbf{y},$$

implying $\mathbf{x}' \sim_{AFW} \mathbf{y}$. Hence, since $\mathbf{x}'$ and $\mathbf{y}$ have the same median, we have $\mathbf{x}' = \mathbf{y}$, implying $\mathbf{x}' \sim \mathbf{y}$ (by reflexivity of $\succsim$). Therefore, by TI, we have

$$\mathbf{x} \sim \mathbf{x}' \sim \mathbf{y},$$

implying $\mathbf{x} \sim \mathbf{y}$.

If $\mathbf{x} \succ_{AFW} \mathbf{y}$, then

$$\mathbf{x}' \sim_{AFW} \mathbf{x} \succ_{AFW} \mathbf{y},$$

implying $\mathbf{x}' \succ_{AB} \mathbf{y}$. Consequently, because $\mathbf{x}'$ and $\mathbf{y}$ have the same median, Theorem 2.1 in Chakravarty (2015) implies that $\mathbf{x}'$ can be obtained from $\mathbf{y}$ by a finite sequence of IS and/or IB transformations, and so $\mathbf{x}' \succ \mathbf{y}$.²⁴ Therefore, by TI, we have

$$\mathbf{x} \sim \mathbf{x}' \succ \mathbf{y}.$$

Since $\succsim$ is reflexive and transitive, this yields $\mathbf{x} \succ \mathbf{y}$. ■

Lemma 6. *For $u \in \mathcal{U}$, $\mathcal{F}_{(\succsim_{AFW}, u)-r} = \mathcal{F}_{(\succ'_{AL}, u)-r}$.*

²³The relation $\sim$ is transitive because $\succsim$ is reflexive and transitive.

²⁴An IS transformation is a (weakly) spread-increasing movement of income levels away from the median income, a movement consistent with the antecedent in IS. An IB transformation results from bipolarity-increasing progressive transfers on either side of the median income, a transformation consistent with the antecedent in IB.

Proof. $\left[\mathcal{T}_{(\succ_{AFW}, u)-r} \subseteq \mathcal{T}_{(\succ'_{AL}, u)-r} \right]$ Take $T \in \mathcal{T}_{(\succ_{AFW}, u)-r}$ and $\mathbf{a} = (a_1, \dots, a_n) \in \mathcal{A}_n$. Choose any pair a_i and a_j with $a_i < a_j$. Let $\mathbf{a}' = (a'_1, \dots, a'_n) \in \mathcal{A}_n$ satisfy $a'_{m-1} := a_i < a'_m < a_j := a'_{m+1}$. Because $T \in \mathcal{T}_{(\succ_{AFW}, u)-r}$, we have

$$y^u(a'_m, 0) - y^u(a'_{m-1}, 0) \geq x^u(a'_m, T) - x^u(a'_{m-1}, T)$$

and

$$y^u(a'_{m+1}, 0) - y^u(a'_m, 0) \geq x^u(a'_{m+1}, T) - x^u(a'_m, T).$$

Arranging terms and using $a'_{m-1} = a_i$ and $a'_{m+1} = a_j$ yields

$$y^u(a_j, 0) - x^u(a_j, T) \geq y^u(a'_m, 0) - x^u(a'_m, T) \geq y^u(a_i, 0) - x^u(a_i, T).$$

Since a_i and a_j were arbitrary coordinates in $\mathbf{a}$, it follows from [Lemma 1](#) that $T \in \mathcal{T}_{(\succ'_{AL}, u)-r}$, as we sought.

$\left[\mathcal{T}_{(\succ'_{AL}, u)-r} \subseteq \mathcal{T}_{(\succ_{AFW}, u)-r} \right]$ Take $T \in \mathcal{T}_{(\succ'_{AL}, u)-r}$ and $\mathbf{a} = (a_1, \dots, a_n) \in \mathcal{A}_n$. Define

$$b_i := y^u(a_i, 0) - x^u(a_i, T).$$

Because $T \in \mathcal{T}_{(\succ'_{AL}, u)-r}$, it follows from [Lemma 1](#) that $b_i \leq b_j$ for all $i < j$. Observe that we can write, for each i ,

$$x^u(a_m, T) - x^u(a_i, T) = y^u(a_m, 0) - b_m - y^u(a_i, 0) + b_i = y^u(a_m, 0) - y^u(a_i, 0) + b_i - b_m. \quad (35)$$

Consider first the case where $i < m$. Then $b_i - b_m \leq 0$. Consequently, from (35) and the monotonicity of x^u and y^u in a ([Lemma 1](#) in [Carbonell-Nicolau and Llavador, 2021b](#)), one obtains

$$x^u(a_m, T) - x^u(a_i, T) \leq y^u(a_m, 0) - y^u(a_i, 0). \quad (36)$$

Similarly, for $i > m$, $b_i - b_m \geq 0$ and

$$x^u(a_i, T) - x^u(a_m, T) \leq y^u(a_i, 0) - y^u(a_m, 0). \quad (37)$$

From (36) and (37), it follows that

$$(y^u(a_1, 0), \dots, y^u(a_n, 0)) \succ_{AFW} (x^u(a_1, T), \dots, x^u(a_n, T)).$$

Since $\mathbf{a}$ was arbitrary in $\mathcal{A}_n$, we conclude that $T \in \mathcal{T}_{(\succ_{AFW}, u)-r}$. ■

E Proofs of [Lemma 2](#) and [Lemma 3](#)

Lemma 2. For $u \in \mathcal{U}$, $b \geq 0$, and $R \subseteq [0, 1)$, $\mathcal{T}_{m\text{-}prog}(b, R) \subseteq \mathcal{T}_{(\succ'_{AL}, u)-r} \Leftrightarrow \mathcal{T}_{lin}(b, R) \subseteq \mathcal{T}_{(\succ'_{AL}, u)-r}$.

Proof. The ‘only if’ part is trivial, since $\mathcal{T}_{lin}(b, R) \subseteq \mathcal{T}_{m\text{-}prog}(b, R)$.

To see that $\mathcal{T}_{lin}(b, R) \subseteq \mathcal{T}_{(\succ'_{AL}, u)-r}$ implies $\mathcal{T}_{m\text{-}prog}(b, R) \subseteq \mathcal{T}_{(\succ'_{AL}, u)-r}$, fix $T \in \mathcal{T}_{m\text{-}prog}(b, R)$. By [Lemma 1](#), it suffices to show that

$$y^u(a', 0) - x^u(a', T) \geq y^u(a, 0) - x^u(a, T), \quad \text{whenever } a' > a > 0,$$

for any pre-tax and post-tax income functions x^u and y^u .

Because $T \in \mathcal{T}_{m\text{-}prog}(b, R)$, there exist a finite partition of $\mathbb{R}_{++}$,

$$A_1 := (0, a_1], A_2 := [a_1, a_2], \dots, A_{K+1} := [a_K, \infty),$$

and $K + 1$ linear tax schedules in $\mathcal{T}_{lin}(b, R)$, $T_1, \dots, T_{K+1}$, such that

$$x^u(a, T) = x^u(a, T_k), \quad a \in A_k, \quad k \in \{1, \dots, K + 1\}.$$

Since, by assumption, the tax schedules $T_1, \dots, T_{K+1}$ are $(\succ'_{AL}, u)$ -r, **Lemma 1** gives, for every $k \in \{1, \dots, K + 1\}$

$$y^u(a', 0) - x^u(a', T_k) \geq y^u(a, 0) - x^u(a, T_k), \quad \text{whenever } a' > a \text{ in } A_k,$$

for any pre-tax and post-tax income functions x^u and y^u .

For $a' > a > 0$, we have $a' \in A_{k'}$ and $a \in A_k$ for some $k' > k$ and

$$\begin{aligned} y^u(a', 0) - x^u(a', T) &= y^u(a', 0) - x^u(a', T_{k'}) \\ &\geq y^u(a_{k'-1}, 0) - x^u(a_{k'-1}, T_{k'}) \\ &= y^u(a_{k'-1}, 0) - x^u(a_{k'-1}, T_{k'-1}) \\ &\geq y^u(a_{k'-2}, 0) - x^u(a_{k'-2}, T_{k'-1}) \\ &\quad \vdots \\ &\geq y^u(a_k, 0) - x^u(a_k, T_k) \\ &\geq y^u(a, 0) - x^u(a, T_k) \\ &= y^u(a, 0) - x^u(a, T), \end{aligned}$$

as we sought. ■

Lemma 3. For $u \in \mathcal{U}$, $b \geq 0$, and $R \subseteq [0, 1)$, $\mathcal{T}_{lin}(b, R) \subseteq \mathcal{T}_{(\succ'_{AL}, u)\text{-r}} \Leftrightarrow u \in \mathcal{U}_\xi(b, R)$.

Proof. By **Lemma 1**, the linear tax schedules $T(y) = -b + ry$ in $\mathcal{T}_{lin}(b, R)$ are $(\succ'_{AL}, u)$ -r if and only if the map

$$a \mapsto y^u(a, 0) - x^u(a, T) = al^u(a, 0) - [a(1-r)l^u((1-r)a, b') + b']$$

defined on $\mathbb{R}_{++}$ is nondecreasing for every $(b', r) \in [b, \infty) \times R$. Equivalently,

$$l^u(a', 0) + a' \frac{\partial l^u(a', 0)}{\partial a} \geq (1-r) \left[l^u((1-r)a', b') + a'(1-r) \frac{\partial l^u((1-r)a', b')}{\partial a} \right]$$

for every $(a', b', r) \in \mathbb{R}_{++} \times [b, \infty) \times R$, or

$$al^u(a, 0)(1 + \xi^u(a, 0)) \geq (1-r)al^u((1-r)a, b')(1 + \xi^u((1-r)a, b'))$$

for every $(a, b', r) \in \mathbb{R}_{++} \times [b, \infty) \times R$. Letting

$$y^u(a, 0) = al^u(a, 0) \quad \text{and} \quad y^u((1-r)a, b') = (1-r)al^u((1-r)a, b')$$

yields (23), i.e., $u \in \mathcal{U}_\xi(b, R)$. ■

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